Faithful attribution recovers the causal wiring, not the semantics: a ground-truth audit of interpretability on a differentiable microprocessor

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Abstract

Interpretability methods are judged by their faithfulness: whether they name the true causes of a system's output. The field treats faithfulness as the test that separates a real explanation from a merely plausible one. We test this on the Understanding VCS Project, a bit-exact and differentiable copy of the Atari 2600 Video Computer System (VCS). The machine was never built to be interpretable. Yet it is fully known, so the true cause of any output can be computed by exact intervention. We score thirty interpretability methods for faithfulness against this known answer (**1** = exact, **0** = chance), aggregated over 44 games. Causal and intervention methods are more faithful than gradient and correlational ones (**0.71** versus **0.38**). The clearest case is a sprite's position, where the naive gradient is provably zero: a causal method scores **1.000** and standard saliency **0.092**. On this position regime the causal advantage is now statistically established: the family-mean gap is **0.34**, with a 95 percent bootstrap interval over the 44 games of **0.29** to **0.38** that excludes zero. But faithfulness is not what is missing. When we switch on the port's differentiable sampler, the naive gradient becomes non-zero—saliency rises from **0.000** to **0.791** on Pong—yet its faithfulness stays low on most games and no method recovers what a variable means. Every method reports which signals matter, never what they mean. We name this remaining distance the semantic gap, and we explore it in detail. Meaning is not in the mechanism. For software it is fixed outside the code, in the documentation and the data the program was written for. No method that reads only the mechanism can reach it. The Understanding VCS Project offers a reusable benchmark with known answers, for explainable AI, mechanistic interpretability, and software verification.

Keywords: interpretability, explainable AI, mechanistic interpretability, faithfulness, ground truth, causal attribution, semantic gap, Atari VCS

Interpretability now matters in practice. Neural networks drive cars, read scans, and approve loans. Before we trust such a decision, we ask why. Yet on the systems we most want to explain we cannot check whether the answer is correct. A saliency map (an input-importance heatmap) for a deep network can look convincing and can change with the action, and still not be a correct explanation [1]. The field of explainable AI has converged on *faithfulness* as its quality criterion: whether an explanation names a system's *true* causes [2]. The difficulty is that on a real neural network there is nothing to measure it against. Sanity checks can show that a method fails, but with no reference map they cannot certify that it is correct [3].

This blind spot is not specific to deep learning. Jonas and Kording applied standard neuroscience methods to a microprocessor in which every transistor is known [4]. Lesions, tuning curves, and connectomics produced rich structure, but none of it recovered how the chip computes. The same gap recurs across the three interpretability traditions that rarely cite one another. The neuroscience analyses record and perturb a system. The attribution tradition asks which inputs mattered for a decision, through gradient and saliency maps [5–8], class-activation methods [9, 10], occlusion [11–13], and surrogate or game-theoretic attributions [14, 15], applied to Atari agents to visualise what a policy attends [1, 16]. Mechanistic interpretability aims past attribution at the circuit, through path patching and causal mediation [17–20], alignment search and interchange interventions [21, 22], and the sparse dictionaries and probes that read features off the state [23–27]. Across all three, methods are validated against each other or against human intuition rather than against the mechanism [28], and even the sparse-dictionary branch reports that its tuning proxies do not track interpretability [29]. Faithfulness has stood in for understanding, and no one could test the stand-in.

Maier et al. recently built the missing reference, in a companion study. It is a bit-exact, differentiable port of the Atari 2600 Video Computer System (VCS) that reproduces a real 1977 machine to the byte and pixel [30, 31]. The VCS is software with a known specification, so the true cause of any output is computable by exact intervention. This *intervention oracle* (Section 3.2) returns the true causal effect for any candidate cause of any output, measured rather than estimated. It operates over the platform shown in Fig. 1. On this machine an interpretability claim can be scored right or wrong. We operationalise interpretability as ground-truth recovery: an explanation is interpretable to the degree that its claimed structure matches the system's true causal structure, scored by the oracle.

We audit 30 interpretability methods, plus the oracle as a positive control (the ground truth scored as if it were a method, so it reaches 1 by construction), on two axes: faithfulness against the oracle, and a documented plausibility proxy for how convincing a method looks (a fixed per-tradition rubric, not a human study). The first result is the expected one, and we treat it as a step rather than the finding: causal and intervention methods are faithful, while gradient and correlational methods are

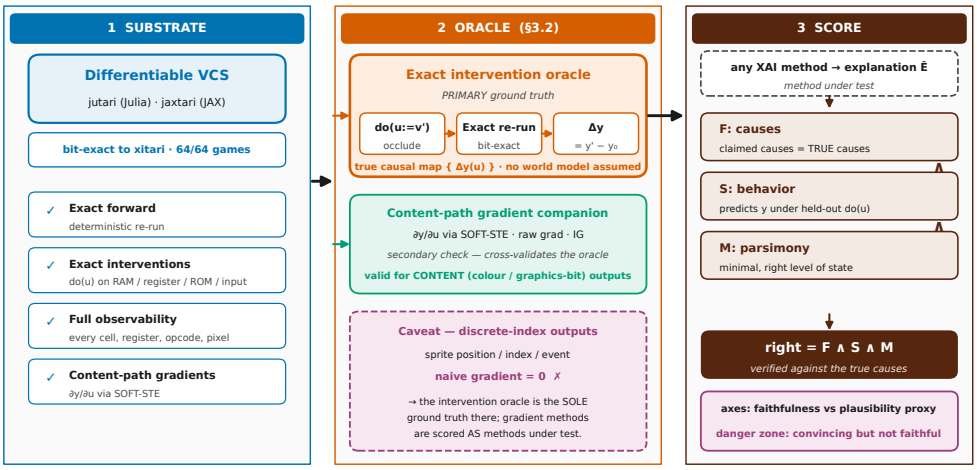


Fig. 1 The measurement platform and its ground-truth oracle. A differentiable, bit-exact port of the Atari VCS (left), realized as two cross-checked implementations, `jutari` (Julia) and `jaxtari` (JAX), gives four capabilities no neural network grants for free: exact forward, exact interventions do(u) (force a cause to a value and re-run), full state observability, and content-path gradients (through the smoothly varying values, not the discrete position read). On it we build the oracle (centre): the primary intervention oracle of Section 3.2, do($u{:=}v'$) $\to \Delta y(u)$ by bit-exact re-run, with a content-path gradient $\partial y/\partial u$ as a companion on the content path only. Any explanation $\hat{E}$ is scored by the correctness triad (right), right = F $\wedge$ S $\wedge$ M, and reported on two axes, faithfulness against the oracle versus a plausibility proxy. Faithfulness aggregates correlation, sign and magnitude agreement, deletion/insertion area-under-curve, and precision at k against the true causal top-k (Section 3.4). A discrete index — a sprite's position, set by strobe timing — has a naive gradient that is zero under the hard forward map (Prop. 1); for position, index, and event outputs the intervention oracle is the *sole* ground truth and gradient methods are scored as methods under test. The schematic draws no experiment data. Data and oracle: see Supplement; oracle defined in Section 3.2.

not, most sharply on the discrete sprite-position outputs whose naive (unsmoothed) gradient is provably zero (Prop. 1). A reader could dismiss that zero as an artifact of a discrete index, so we remove it: turning the differentiable port's bilinear sampler on — a sub-pixel interpolation that makes the discrete position read smooth — restores the position gradient, and the gradient methods become faithful too on the games with a moving tracked sprite. A gradient through the whole running program, instead of a single read, returns the same map on most games and no better one on average. Making them faithful does not change the conclusion, because faithfulness was never the dividing line.

With the same oracle we ask what a faithful method *delivers*, and the answer is always the same kind of object: a causal-effect map or a data-flow graph, never the variable's meaning or the routine's algorithm. We call an explanation right only if it is **F**aithful (its claimed causes are the true ones), **S**ufficient (it regenerates behavior under held-out interventions), and **M**inimal at the algorithmic level — the level of the function computed, not its wiring — (right = $\mathbf{F} \wedge \mathbf{S} \wedge \mathbf{M}$, Section 3.4). The data then show a gap in both directions: a circuit that is graph-correct and minimal yet

does not reproduce the behavior it explains, and a learned feature that matches a named game variable perfectly yet is causally unused: removing it does not change the output. We name this residual the *semantic gap*, and it does not shrink where faithfulness is restored. The per-method scores are reported in Section 1 (Tables 1–3 and Figs. 2 and 3).

The reference for meaning lies outside the mechanism. Every semantic label in the study entered from outside the system, imported from external annotation sets [32, 33] and only *causally verified* on our substrate (the differentiable VCS port), never *discovered* by a method under test. An alignment method that reports perfect accuracy is therefore right by construction, because it checks a variable we supplied. Probing shows that information is present, not that it is used [25], and our oracle is the strictly stronger test. What fixes the meaning is the system's *behavior*. For software, that behavior is set down in its documentation and data: the manuals, the specification, and the design. IEEE 610.12, the software-engineering glossary, makes these part of the definition of software itself [34]. The wiring is in the mechanism, so any faithful method can recover it. The meaning is not, and no method we tested supplied it.

To our knowledge this is the first cross-tradition audit of interpretability methods on a real, human-written software artifact whose ground truth is independent of it, scored by an exact intervention oracle; we claim no unqualified "first," since synthetic known-answer benchmarks and Jonas and Kording's microprocessor study [4] precede us. We read a 1977 chip as a necessary-condition screen, not a predictor of success on learned networks: under perfect ground truth and no learning, failure here is strong negative evidence for harder, less observable systems, and the chip still shows the properties that make interpretation hard, from aliased RAM cells to race-the-beam timing (Section 2). We are explicit about the limits: no method recovered the program's meaning, algorithm, or design from scratch, and the plausibility axis is a documented proxy, not a human study. This paper *defines and measures* the semantic gap and locates its reference in the system's behavior; *closing* it is future work, hopefully inspired by this paper. The hard part of interpretability is not faithfulness but recovering meaning, which remains unsolved even when faithfulness is achieved. We believe that naming and demonstrating the existence of this gap is a first step toward closing it.

1 Results

1.1 The neuroscience analyses

We begin with a known concern in the field. Jonas and Kording asked whether a neuroscientist with the standard toolkit could understand a microprocessor. They found that the toolkit yields rich structure but misses the mechanism [4]. They could show the failure but not measure it. On the 6502 (the CPU model) they had no causal ground truth to score each method against. We supply that missing number. We run the same analyses on the bit-exact, fully known VCS, where exact intervention recovers the true causal role of every register and RAM cell (Section 3). This makes Kording's qualitative lesson quantitative. Across the eight analyses A1–A8, faithfulness against the oracle ranges from 0.16 to 1.0 and averages 0.46 [35]. The oracle-as-method positive

control sits at 1.0 by construction (Table 1). The structure these methods report is real. But structure is not the same as cause. The score measures the difference.

The dependency structure is only partly recoverable. A1 reconstructs the dependency graph over state variables, a connectome of cell-to-cell edges found by single-shot perturbation, and scores it against the true data-flow graph at a faithfulness of 0.21 [35]. Pooled over the scored games, 46 of the 60 edges it reports are real, but it finds only 46 of the 409 true edges. The single-unit lesion A2 fares best. It clamps each RAM cell in turn and ranks it by the behavioral damage it does. The resulting importance map has a Spearman correlation of about 1.0 with the unit's true causal role on the six core games (listed in Section 1.2), and averages 0.96 over the scored games [35]. Lesioning is causal, which is why A2 is the one classical method that scores near the top.

Yet a near-perfect rank order still misses interactions. A single-unit lesion changes one cell at a time, so it is blind to causes that act only together. On Space Invaders the per-unit ranking misses 15% of the pairwise interactions the oracle records (8 of 55 pairs; 19% averaged over the scored games). One super-additive pair moves the importance score by 40 beyond the sum of its parts [35]. The ranking of which units matter is correct, but the model of how they combine is missing. So even the strongest classical method recovers a ranking, not the computation. An effect map does not show the algorithm behind it.

Tuning curves report structure that does not match the true cause. A3 fits per-unit tuning to a game variable, the workhorse of systems neuroscience. On the VCS it actively misleads. Its *spurious-tuning rate* is the fraction of strongly-tuned units whose tuning does not match their true causal role. That rate is 0.50 on Pong. Its faithfulness, the rank correlation between tuning selectivity and the oracle's causal importance, averages 0.21, among the lowest of the analyses [35]. Tuning is correlational. On a machine where many cells co-vary with the frame clock, apparent tuning is cheap. This is the same present-versus-used distinction we discuss later for linear probing.

Correlation structure and field potentials are real, but they are mostly side effects, not causes. A4, the spike-word analysis (which cells fire together) of pairwise and global correlation, reproduces the familiar weak-pairwise/strong-global signature on 24 of its 36 scored games. Yet against the true coupling its faithfulness is only 0.25, with low sufficiency ($S = 0.15$; S is a held-out predictive fit, which can be negative per game and is clipped to $[0, 1]$ before averaging) [35]. A5 is the local-field-potential (voltage from nearby neurons) analogue that pools regional activity into power spectra. It scores 0.39, and much of that variance is the known clocks, the frame counter and the free-running timer cells [35]. Both methods detect the machine's activity but not its computation.

Granger causality treats "happens earlier" as "is the cause." A6 infers directed edges between RAM cells and between the CPU, TIA, and RIOT (the TIA is the Television Interface Adaptor, the graphics/sound chip, and the RIOT the RAM-I/O-Timer). Its faithfulness is the F_1 of the inferred cell-to-cell edges against the true data-flow. On Pong its faithfulness is 0.0, with false-edge and missed-edge rates both at 1.0 [35]. The VCS is clock-locked, so nearly everything is predictable from everything else one cycle earlier. Equating precedence with causation therefore infers mostly

Table 1 The neuroscience analyses scored against the intervention oracle. Faithfulness F of each classical analysis A1–A8 against the exact intervention oracle (Section 3.2), with its 95% bootstrap confidence interval over the scored games (36–44 per analysis) and the documented plausibility proxy (Section 3). The analyses average 0.46; the oracle-as-method positive control is 1.0 by construction. Only the causal single-unit lesion (A2) reaches the ceiling on a non-trivial metric. The whole-state recording (A8) is at 1.0 only because recording everything is faithful, yet its minimality is 0.07 — a log file, not an account. Every value reads from a committed per-game record via the leaderboard [35]; none is re-run or estimated.

Analysis	F (95% CI)	Plaus	What F scores
A1 connectomics	0.21 [0.13, 0.29]	0.55	data-flow-graph F_1 vs. oracle
A2 single-unit lesions	0.96 [0.93, 0.99]	0.55	lesion-importance ρ vs. true role
A3 tuning curves	0.21 [0.14, 0.29]	0.80	tuning selectivity vs. causal importance (ρ)
A4 spike-word correlations	0.25 [0.19, 0.31]	0.80	coupling vs. true coupling
A5 local field potentials	0.39 [0.34, 0.45]	0.80	1−clock-explained variance
A6 Granger causality	0.16 [0.11, 0.20]	0.80	directed-edge F_1 vs. data-flow
A7 dimensionality reduction	0.52 [0.47, 0.58]	0.75	NMF matched-component fraction
A8 whole-state recording	1.00 [1.00, 1.00]	0.30	complete record (minimality $M = 0.07$)
Oracle (positive control)	1.00 [1.00, 1.00]	1.00	exact intervention ground truth

spurious edges: averaged over the scored games, 77% of the inferred edges are false and 76% of the true edges are missed. We ran one setting (lag 2, $\alpha = 0.05$) and did not sweep lag or threshold. Precedence and causation differ here, and only the intervention oracle separates them.

Dimensionality reduction scores in the middle. A7 decomposes the state tensor with NMF (non-negative matrix factorization) and PCA (principal component analysis) and matches the top components to the known game variables and the known clock and read/write signals. Its score is the matched fraction, the share of the top five components that align with a known variable or signal. On the core games both factorizations match about 60 percent (NMF 0.60, PCA 0.57). Over the scored games NMF averages 0.52 and PCA only 0.34; the score we report is NMF's [35]. This makes A7 the highest score after the causal lesion and the trivial whole-state record. It still leaves about half of the components unmatched.

A complete record is still not an explanation. A8 records the full RAM and register state, faithful and sufficient by definition ($F = 1.0$, $S = 1.0$; faithfulness $F \in [0, 1]$ runs from chance at 0 to the oracle ceiling at 1). Yet its minimality is only 0.07 ($M \in [0, 1]$) [35]. At the algorithmic level, a recording of every cell explains nothing. Faithfulness and sufficiency without minimality give a log file, not an account of the computation. This is the first sign that the three axes are independent. The semantic-gap result makes that point central later.

Taken together, these analyses confirm the expected result and puts a number on it. The advance over Jonas and Kording is that we grade the *architectural* register-transfer level (registers and the logic between them) with a *quantitative* F against an exact oracle, rather than giving a qualitative verdict on the transistor. The contrast is clear, because the causal operators we apply to the same machine in Section 1.2 and Section 1.3 reach the ceiling these analyses cannot. So the limitation lies in the analysis methods, not in the machine. We mean this strictly as calibration. Even the best of these analyses recover a ranking, a graph, or a record, never the algorithm the program runs. This remaining gap is what the paper measures.

1.2 Attribution: which inputs caused the output

Attribution returns a map of which inputs mattered. We apply these methods to the VCS to ask a question a real network cannot answer: is the map true? Every attribution is scored against the same intervention oracle (Section 3.2) that defines ground truth. For each chosen output y, a pixel, a score, or a game event, we know the exact set of causes it should have named. We ran twelve attribution methods across the 44 scored games, with the six core games (Pong, Breakout, Space Invaders, Seaquest, Ms. Pac-Man, Q*bert) as running examples, plus an oracle-as-method positive control, and score each against the intervention oracle in Table 2. We score all methods from a shared live state rather than a boot frame. We seed a single random-action prefix of 90 steps and a horizon of 15 (seed 0), and check the state with the oracle's cause-density gate: at least 4 candidate causes must move the position output (below) by more than 0.5. The gate flags a state; it does not reject it. All 44 scored games pass. Every method explains the same two outputs per game (Section 3). The content output is one RAM byte at the end of the horizon. The position output is the number of screen pixels, over the whole frame, that differ from the un-intervened frame; the gradient methods take its derivative through the bilinear sampler at one sprite cell. The live state matters: on Seaquest the old boot-frame content oracle moved only 2 of 48 candidate causes, while at the gameplay state 19 of 48 move the output above the floor, so the methods are scored where the program is actually computing.

The main result is a split between two output types, not a ranking of methods. A content output is the value of a register, a colour, or a graphics bit. It is a differentiable function of the state, so its gradient flows. A position output is where a sprite is drawn, set by strobe timing on a discrete index. It has provably zero naive gradient (Paper 1, and Section 3). Hence the whole gradient family hits this limit, and it splits the leaderboard into two clear groups.

On the content path the gradient is faithful, as expected. Vanilla saliency [5] puts most of its attribution on the true causal byte: on Q*bert its attribution of a content read correlates with the oracle at Pearson 0.999 with precision@3 of 1.0, on Ms. Pac-Man at 0.811 / 1.0, on Breakout at 0.750 / 1.0. Integrated Gradients (IG) [6], Grad×Input/DeepLIFT [36], SmoothGrad [7], and guided backpropagation [8] all reproduce this, because the output we explain is a one-hot read whose Jacobian $\partial y/\partial u$ is a constant pointing at the read index. Smoothing does not change the score (Δ corr ≈ 0), and guided backprop reduces to vanilla saliency. On this path all methods give the same answer.

The gradient family is weak on the position output. The naive gradient there is identically zero, so the raw saliency, Grad×Input, SmoothGrad, guided-backprop, and IG maps carry no signal through the position read (Prop. 1). A differentiable substrate can restore a non-zero gradient here. The battery scores the position output with that repair, the bilinear sampler, switched on (Section 1.4), but the restored gradient is faithful only on a minority of games and its faithfulness stays low. Averaged over the position regime the gradient and correlational family reaches only 0.22 against the oracle, against 0.56 for the causal and intervention family (Table 2). Across all output regimes the gradient and correlational family scores 0.38 and the causal and

intervention family 0.71. The position map is not empty by choice; it is weak exactly where the discrete game logic sits.

One objection remains: these gradients differentiate a single read at the intervention frame, not the program that runs during the horizon. Vanilla saliency, Grad×Input, IG, Expected Gradients, SmoothGrad and guided backprop are in that sense local read-out surrogates of the true gradient. We therefore ran a true end-to-end control on all 44 scored games. We read the output at the end of the horizon and differentiate it back to the RAM at the intervention frame through every CPU instruction in between, 57,073 to 106,547 instructions per game. The pass is checkpointed every 200 instructions, and the forward pass is bit-exact to the hard machine in every game. TIA and RIOT reads enter as constants, so the renderer itself is not differentiated. On most games the result is the same map. On the content output the end-to-end per-cause map equals the single read's on 35 of the 44 games. On 33 of them the gradient is exactly e_{idx} ($\|g\|_1 = 1$, one non-zero entry). On Pong and Atlantis it also reaches a byte outside the candidate causes with $\partial y/\partial u = 15$, the horizon length, which is what a per-frame velocity added to a position produces. On 8 games the gradient vanishes, because the straight-through gradient finds no arithmetic path from the intervention frame to the byte at the end of the horizon. This costs Enduro (1.000 to 0), Phoenix (0.909 to 0) and Air Raid (0.360 to 0) their content score. One game gains: in Centipede the content byte at the end of the horizon is carried over from `ram[111]` at the intervention frame (unit derivative, no other support), the gradient follows it to the oracle's largest cause, and its faithfulness rises from -0.096 to 0.851. On the position output the map is unchanged on 35 games and vanishes on 8, among them Breakout and Space Invaders, although the sampler's gradient at the end of the horizon is non-zero. Breakout recomputes its paddle position every frame from the paddle input, a hardware read the gradient cannot see. Averaged over the games, faithfulness falls from 0.631 to 0.598 on content and from 0.092 to 0.086 on position. Differentiating the whole program therefore does not rescue the gradient family. The gradient follows data that is copied or added. Every other path by which the oracle's causes reach the output runs through a branch, an indexed load, or a hardware read, and the straight-through gradient carries none of these. It sees the same smooth sliver of the computation.

Two further results make this point stronger. The baseline choice is the perennial worry about IG. Here it changes magnitude, not faithfulness. The $(x - x_0)$ factor moves $\max|\text{IG}|$, yet the ranking and the oracle correlation hold, because the constant one-hot Jacobian keeps the mass on the same byte. Only the degenerate baseline equal to the content byte breaks it. Expected Gradients [37] averages IG over a distribution of recorded VCS states. It removes that degeneracy and is stable across reference draws: over five reference seeds its oracle correlation varies by less than 10^{-15} on every scored game. On 20 of the 44 games, however, its content attribution is identically zero, mostly because the content byte holds the same value in every reference state, so every $(x - x_0)$ factor vanishes. On the content path it matches a single-baseline IG on 25 games, falls below it on 17 (16 by more than 0.05) and exceeds it on 2 (by at most 0.04), at $K\times$ the compute (for K baseline draws), and it still vanishes on position.

The more careful method gives stability but not accuracy. Its all-regime faithfulness of 0.19 is the lowest among the attribution methods.

Guided backprop and vanilla saliency fail the program-randomization sanity check of Adebayo et al. [3] on the system itself. We scramble the ROM so the program boots to a completely different state (on average 70–127 of 128 RAM bytes change, lowest on Seaquest), and the content-path saliency map stays byte-identical (cosine ≈ 1.0 on all six games). The Jacobian $\partial y/\partial u = e_{\text{idx}}$ is a constant one-hot, independent of what the program computes. So the map points at the read index no matter which program produced the output. The saliency is therefore provably model-invariant. A faithful explanation must change when the model changes. This one does not. The intervention oracle, by contrast, distinguishes the two programs at once. The map identifies the right cause but does not depend on the model that produced it.

When we replace the gradient with a real intervention, the position result improves. Occlusion [11] sets a candidate cause and re-runs the bit-exact program, a coarse version of the oracle. On the position regime, where the naive gradient is zero, it reaches faithfulness 0.619, and its all-regime faithfulness of 0.685 is the highest among the attribution methods. The black-box perturb-and-re-run methods follow the same logic and result: KernelSHAP / Shapley sampling [15, 38] (0.650), LIME [14] (0.640), and RISE [13] (0.544) each draw masked coalitions (random held-out subsets of candidate causes) and average over the genuine responses, so each works on position outputs where the gradient is dead, and each reports its own completeness ($\Sigma\phi = y(\text{intact}) - y(\text{all-masked})$ for KernelSHAP, LIME's surrogate R^2, the fit of its local linear model). The leaderboard nevertheless tags these three as attribution-tradition methods and counts them in the gradient and correlational family, not in the causal and intervention family.

The two minimal-mask methods meet Atrey et al.'s [28] objection that perturbations stray off-manifold (a byte value the running program would never itself write). Extremal perturbation [12, 39] optimises an area-bounded mask of real *do*(occlude) re-runs, and the on-distribution counterfactual [40] substitutes candidate cells toward a genuine alternative VCS state, the RAM of another frame of the same ROM. Both stay on the data manifold by construction and both work on position outputs (position-regime faithfulness 0.34 and 0.28, all-regime 0.46 and 0.43). The counterfactual has no effect (a no-op) on some games, for example where no on-distribution content byte varies between frames. Its map is then flat, and these cells enter its mean as $F = 0$; we do not drop them. The cross-tradition contrast is clearest across all three phases: the causal and intervention family averages 0.71 faithfulness [$n = 11$ methods] against the oracle, against the gradient and correlational family's 0.38 [$n = 14$] (Table 2). Restricted to the attribution methods alone the two means are closer, because the content-path gradient is faithful and lifts the gradient mean. The intervention advantage is a cross-tradition and position-regime effect, not an attribution-wide one, and we report it as such. A perturbation is faithful only when it is a valid intervention.

Some popular methods return no answer at all, because the VCS has no structure for them to use. Grad-CAM and Grad-CAM++ [9, 10] need a final convolutional layer to weight; attention rollout [41] needs attention matrices; VIPER [42] needs a learned policy to distil. The VCS is a 6507 CPU (a reduced-pin 6502) with a TIA and a RIOT

executing a fixed ROM. Across all six games the count of each required substrate is exactly zero, against ≥ 16 genuine candidate causes per game for the applicable methods. We record this as a measured finding, `applies` = `false`, not a poor score: the absence is structural and the same in every game, and re-targeting Grad-CAM's pooling onto raw register state would only be the content-path gradient renamed. Several well-known methods produce no result on a real artifact. This absence is itself a finding of the audit.

Across the twelve methods, faithfulness and plausibility move in opposite directions: methods rank high on plausibility but low on faithfulness. The clearest pair makes it concrete. On the position regime the causal method activation patching (Section 1.3) reaches faithfulness 1.000, the oracle ceiling, while vanilla saliency, a widely used baseline, reaches only 0.092. This is a per-method position gap of 0.908. Yet, on the plausibility proxy of Section 3 the popular method has the higher score (0.9 versus 0.5): the weaker map has the higher plausibility score. Low faithfulness with high plausibility is the danger zone (Section 3). On the VCS we can measure it directly, not just warn about it.

We claim only that these maps are faithful, nothing more. A correct occlusion map names the true causal bytes. It does not tell us what those bytes mean, that a cell is the ball's x or that a routine is the bounce. Object-level semantics enter only through the external Tier-3 (T3) labels, game-concept names imported from AtariARI and OCAtari (Section 3), and even there we verify a supplied label by intervention rather than discover it. Whether a faithful attribution amounts to understanding the computation we take up in Section 1.4. Here we have established only that faithfulness is achievable, and that the most popular methods do not achieve it.

1.3 Mechanistic interpretability on a known circuit

Mechanistic interpretability makes the strongest claim in the field. It tries to explain how a system computes, not just where it attends. Its methods read internal activations as a circuit and try to recover that circuit. On a neural network the recovered circuit can only be checked against itself, because the true wiring is unknown. The VCS removes that limitation. Its data-flow is the program's data-flow, known exactly from the disassembly. Every intermediate value is a RAM cell or a register we can read, clamp, and re-run bit-exactly. We therefore run ten mechanistic-interpretability methods on a circuit whose answer we already know. We score each against the same Section 3.2 oracle: the exact intervention table, the true read/write graph, and the $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ triad of Section 3. This calibrates the toolkit against a *known* circuit inside a real, human-written artifact rather than a synthetic or compiled one. Table 3 places the mechanistic methods against the oracle. The result splits cleanly. The causal methods are exact, but their exactness recovers the wiring of the circuit without recovering what the variables mean.

Table 2 Attribution scored against the intervention oracle (Section 3.2). Twelve attribution methods aggregated over the 44 scored games (oracle-as-method positive control = 1.0). F is the all-regime faithfulness against ground truth with its 95% bootstrap CI over games; F_{pos} is the position/index regime alone, where the naive gradient is provably zero; *Plaus* is the documented method-tradition plausibility *proxy* (Section 3), *not* a human measurement. The naive gradient is provably zero on the position read (Prop. 1); the small non-zero F_{pos} the gradient methods still carry is the sampler-restored gradient of Section 1.4, with which the battery scores every position output; it is faithful on only a minority of games and stays low. Across all regimes the causal and intervention family leads (0.71 against 0.38, position 0.56 against 0.22), and the popular gradient methods carry the *higher* plausibility tag — the danger zone, measured. The two family-mean rows are the cross-tradition families ($n = 14$ gradient/correlational, $n = 11$ causal/intervention across all three traditions), not the twelve rows above; restricted to attribution alone the two means are closer, because the faithful content gradient lifts the gradient mean. In the *Family* column, *interv.* rows belong to the causal/intervention family; *gradient* and *black-box* (perturb-and-re-run) rows are counted in the gradient/correlational family, following the leaderboard's tradition tags. The on-distribution counterfactual is a no-op on some core games such as Space Invaders, Ms. Pac-Man, and Q*bert (the substitution toward the alternative frame moves nothing, so $\Delta = 0$, the map is flat, and these cells enter its mean as $F = 0$). Grad-CAM, Grad-CAM++, attention rollout, and VIPER [9, 41, 42] do not apply (no convolution, attention, or learned policy) and are omitted. Values read from the committed leaderboard [35].

Method	Family	F (95% CI)	F_{pos}	Plaus
Occlusion [11]	interv.	0.69 [0.63, 0.74]	0.62	0.55
KernelSHAP [15]	black-box	0.65 [0.59, 0.71]	0.59	0.90
LIME [14]	black-box	0.64 [0.58, 0.70]	0.58	0.90
RISE [13]	black-box	0.54 [0.50, 0.59]	0.43	0.90
Grad×Input / DeepLIFT [36]	gradient	0.47 [0.44, 0.51]	0.09	0.90
Extremal perturbation [39]	interv.	0.46 [0.41, 0.51]	0.34	0.55
On-distribution counterfactual [40]	interv.	0.43 [0.35, 0.51]	0.28	0.55
SmoothGrad [7]	gradient	0.36 [0.30, 0.42]	0.09	0.90
Vanilla saliency [5]	gradient	0.36 [0.30, 0.42]	0.09	0.90
Integrated Gradients [6]	gradient	0.36 [0.30, 0.42]	0.09	0.90
Guided backprop [8]	gradient	0.32 [0.26, 0.37]	0.00	0.90
Expected Gradients [37]	gradient	0.19 [0.13, 0.25]	0.03	0.90
Gradient/corr. mean		0.38	0.22	—
Causal/interv. mean		0.71	0.56	—

Exact recovery of the wiring

The main result is simple. Activation patching recovers the true causal-effect table exactly. Across all six core games the recovered per-site effect equals the exact intervention oracle to the last bit (maximum absolute deviation 0.0), with site precision and recall both 1.0. On this machine, clamping a RAM cell to a donor value and rerunning the real ROM is literally the oracle's own operation. So it cannot disagree with the oracle. Causal scrubbing inherits the same strength. The true data-flow hypothesis preserves behaviour under resample-ablation of everything outside it on all six core games (preserved performance 1.0), and averages 0.997 over the scored games. A deliberately wrong hypothesis does not, and the two are discriminated on every core game. Interchange interventions with distributed alignment search (DAS) verify the supplied alignment at interchange accuracy 1.0 against a mis-aligned control at 0.0. The logit lens reads the true intermediate at readout fidelity 1.0 on every game, yet only 0.79 of its readout weight lands on causally-relevant sites — its faithfulness. Two of the causal methods sit at faithfulness 1.0, level with the oracle positive control

Table 3 Mechanistic interpretability scored against the intervention oracle (Section 3.2). Ten methods aggregated over the scored games (26–44 per method, fewer where a score is undefined; oracle-as-method positive control = 1.0): faithfulness F with its 95% bootstrap CI, the plausibility proxy (Section 3), and the kind of recovery the oracle certifies. The causal core is exact by construction; the approximate and search-based methods lose recall; and the named-but-unused and present-but-unused methods sit in the danger zone (high plausibility, low faithfulness). Values read from the committed leaderboard [35].

Method	F (95% CI)	Plaus	Recovery
Activation patching [17]	1.00 [1.00, 1.00]	0.50	exact effect table
Interchange interventions (DAS, distributed alignment search) [22]	1.00 [1.00, 1.00]	0.50	verifies a supplied alignment
Logit / tuned lens [26]	0.79 [0.73, 0.85]	0.50	exact readout (F = faithful-weight share)
Causal scrubbing [43]	1.00 [0.99, 1.00]	0.50	exact data-flow preserved
ACDC (automatic circuit discovery) [20]	0.65 [0.55, 0.74]	0.50	search (graph-correct, S=0.31)
Path patching [19]	0.60 [0.47, 0.72]	0.50	search (single-shot edges)
Attribution patching [27]	0.48 [0.37, 0.59]	0.90	approximate (linear surrogate)
NMF/PCA dictionaries	0.13 [0.07, 0.19]	0.75	named, partly unused
Sparse autoencoder [23]	0.15 [0.09, 0.21]	0.75	named yet unused (F=0.04 on Pong)
Linear probing (+ control) [25]	0.21 [0.14, 0.30]	0.85	present, not used
Oracle (positive control)	1.00 [1.00, 1.00]	1.00	exact intervention ground truth

(activation patching and DAS), and causal scrubbing at 0.997 just below it (Table 3). Where a method's operation *is* an intervention, faithfulness is automatic on a fully observable machine. This is the first part of the problem, and it is solved here.

Approximate and search-based recovery

The causal account weakens as soon as a method replaces exact re-runs with an approximation or a search. Attribution patching, the gradient-linear surrogate for activation patching, on the six core games correlates with the exact patch at only 0.50 and matches it inside the linear regime in just 0.17 of cases. Yet its recovered edges stay precise (edge precision 0.98, recall 0.70) because the linearisation is wrong about the size of an edge, not about whether the edge exists. Path patching, run in the single-edge style of the IOI (indirect-object-identification) circuit study as a single directed do-operation, recovers the directed edge set of the true routine at mean $F1 = 0.25$ on the core games (0.60 over the scored games). On Q*bert it matches the empty direct circuit perfectly ($F1 = 1.0$). On Pong it recovers none of the eleven true edges ($F1 = 0.0$), because each receiver's value is re-derived within the horizon, so no one-step path carries a surviving signal. ACDC discovers the circuit automatically by resample-ablation edge pruning over a threshold sweep. It reaches mean $F1 = 0.47$ (ROC-AUC 0.70) on the core games, and 0.65 (ROC-AUC 0.78) over its 38 scored games. A positive control re-runs the identical edge-restricted procedure under the exhaustive do-set and scores $F1 = 1.0$ on every game, so the deficit is a real measurement of single-shot recovery, not a broken scorer. These are the methods the field reaches for when the exact intervention is too expensive. On a machine where the exact intervention is cheap, these methods miss real structure. They are precise about what they find but silent about dependencies re-derived through other cells. The limit is recall, not precision. The gap

to the positive control is the quantified cost of approximation against a circuit that is known.

Three separations from the oracle

This subsection presents the central result. ACDC on Q*bert recovers the data-flow circuit at perfect $F1$ against the true routine, and the recovered graph is minimal at the algorithmic level (F= 1.0, M= 1.0). Yet that same graph, scrubbed and asked to regenerate the behaviour under held-out interventions, preserves only S= 0.10 of it (mean scrub-preserved performance 0.31 across games). A circuit can have the correct graph and still not reproduce the computed function. The missing object is not a denser or cleaner graph of the same type. Activation patching has already recovered the effect table exactly, with no tuning budget left to spend. A table of per-site effects is still not the bounce algorithm. One method is exact and the other recovers a perfect graph. Even so, neither reproduces the behaviour. A "methods underperform" explanation cannot account for this separation, because no method underperformed here.

The dictionary methods show the same gap from the opposite direction. A sparse autoencoder (SAE) trained on the recorded RAM trajectory reconstructs the state almost perfectly (fraction of variance explained > 0.99 on Pong). On Pong it matches every varying verified game-variable cell (matched fraction 1.0, 4 of 4; mean 0.88 over its 42 scored games). By the metric the field uses to claim a feature "is" a concept, the SAE succeeds. By the oracle, it does not. Ablating the matched Pong ball feature and re-running the real ROM changes the screen by 0 pixels. The SAE's per-cell recovery correlates with the oracle's causal importance at only F= 0.041, with S= -0.46. When we remove the matched feature, the computation it is supposed to name does not change. Across the scored games the SAE's mean leaderboard faithfulness is 0.15 against a method-tradition plausibility proxy of 0.75, squarely in the Section 3 danger zone. The NMF/PCA dictionary contrast repeats the pattern at lower resolution (matched-component fractions up to 0.75 on the core games beside a causal agreement of only F= 0.13). This separation is the reverse of the first. The circuit was causal but behaviourally incomplete; the feature is perfectly named but never used. The two failures point in opposite directions.

Linear probing shows a third version of the same gap, the well-known probing trap. With control tasks (Hewitt & Liang's selectivity), probes decode game-concept cells well above chance (mean probe accuracy 0.95, mean selectivity 0.10), confirming that the information is present in the state. Yet the program does not necessarily use that information. Cross-referenced against the exact intervention oracle, the oracle flags 14 concept cells across the core games that the program never causally uses within the window (the 15-frame analysis horizon). Of those, 2 are also decodable above chance, so a probe would have nominated them as encoded concepts. Breakout's score cell 84 sits in the larger set: it is present in the state and linearly associated with the displayed score, yet provably unused under intervention within the window. The oracle is strictly stronger than the probe. The probe asks whether a value can be read out. The oracle asks whether changing that value changes the output. Probing earns a high plausibility proxy (0.85) on a faithfulness of 0.21 (the correlation between probe selectivity and the oracle's causal-use label, defined on 26 games), squarely in the danger zone (Table 3).

We state the caveat directly: these labels are imported externally (AtariARI, OCAtari) and only *verified* causally, never *discovered*. So DAS's alignment accuracy of 1.0 is guaranteed in advance, because it only confirms a variable we supplied. The probe tells us a concept is present. Only the intervention tells us the program uses it.

Exactness is necessary but not sufficient

Mechanistic interpretability is the clearest test of the paper's claim. Run against a known real circuit, the mechanistic toolkit splits into two groups. The first is a causal core that is faithful by construction: activation patching and DAS at 1.0, and the logit lens at 0.79, with causal scrubbing at 0.997. Together with the intervention methods this causal and intervention family averages 0.71 faithfulness, against 0.38 for the gradient and correlational family. The second is a recovery layer that is either precise but partial (attribution and path patching, ACDC) or present but unused (SAEs, dictionaries, probing). The oracle, not our judgement, decides the three separations. They appear exactly where the field's success metrics report success: graph-correct yet behaviour-incomplete (ACDC on Q*bert F= 1.0/M= 1.0, S= 0.10), named yet causally unused (SAE on Pong matched 1.0, F= 0.041), present yet unused (cell 84). The faithfulness advantage that the causal and intervention family holds over the gradient methods is robust across all output regimes. The all-regime gap is 0.32 (95% CI $[0.30, 0.35]$, excluding zero). Restricted to the position-output regime alone the gap is 0.34, in the same direction, and its 95% bootstrap interval over the 44 games $[0.29, 0.38]$ now excludes zero, so on position the causal advantage is statistically significant, no longer only directional. The separations above do not rest on either family mean, because each is a within-method contradiction on a single game. On a machine that gives faithfulness for free, faithfulness is necessary for understanding but not sufficient. The residual is the semantic gap: the wiring is recovered, but the meaning is not. Section 1.4 places this gap on the shared axes, and the Discussion treats it as the field's open problem.

1.4 Across the traditions: causal structure without meaning

We have now run three traditions on one machine. This section is organized not by the ranking among them, but by a boundary they share. Every method, faithful or not, returns the system's causal structure but not the meaning of its variables. Faithfulness is whether an explanation names the true causes. On the VCS we can measure it exactly. The intervention oracle of Section 3.2 re-runs the bit-exact program for every candidate cause. It returns ground truth that no real network can provide. That exact answer lets us do two things in sequence. First, we place all three traditions on the shared faithfulness-versus-plausibility plane. We read off the leaderboard and its danger zone, the known result that faithful and convincing are not the same. Second, we go further. We make the unfaithful methods faithful by turning on the differentiable port's bilinear sampler. Making every method faithful does not change the result, because faithfulness was never the dividing line. The harder part of the problem is the variable's meaning and the routine's algorithm. This is a separate object, and no method on this fully known machine recovers it. We call the distance to it the

semantic gap. The leaderboard is an intermediate result; the semantic gap is the main result. The correctness triad we apply throughout — **F**aithful, **S**ufficient, **M**inimal — is defined formally in Section 3.4; here we only read its scores.

The faithfulness leaderboard

The leaderboard separates cleanly along the faithfulness axis. We read it as an exploratory cross-tradition ranking, not a single commensurable score, because the traditions measure different objects. We report the ranking with its robustness. We aggregate the 1862 committed per-game records into the canonical 30 interpretability methods plus the intervention oracle as a positive control, 31 rows in all [35] (Fig. 2). Scored over the 44 games, the causal and intervention methods average a faithfulness of 0.709 against the oracle, while the gradient and correlational methods average 0.384, a gap of 0.324 across all output regimes. This all-regime gap is the robust headline result: its 95% bootstrap interval over games is $[0.30, 0.35]$ and excludes zero [35]. The position regime is the set of discrete sprite-position outputs whose naive gradient is provably zero under the hard forward map (Section 1.2). On this regime the gap points the same way, causal and intervention 0.560 against gradient and correlational 0.222, a gap of 0.338; at 44 games its bootstrap interval $[0.29, 0.38]$ now excludes zero, so on position the causal advantage is statistically significant, not merely directional. The ordering is the one the method matrix predicted (Section 3), now measured. Activation patching and interchange interventions sit at the oracle ceiling of 1.0, causal scrubbing at 0.997, the single-unit lesion at 0.964, and the logit lens at 0.79; the classical correlational analyses (tuning curves, Granger causality, and spike-word correlations) score low, 0.16 to 0.25. The clearest single-method contrast is on the position regime. Activation patching reaches faithfulness 1.000, the oracle ceiling, while vanilla saliency, a widely used attribution baseline, reaches only 0.092, a per-method gap of 0.908 [35]. Hence, on a system where the answer is known exactly, an everyday attribution tool scores far below a causal method on the discrete outputs.

The all-regime separation survives reweighting, and where it does not we say so. Reading the robustness record [35], the causal-versus-gradient gap is 0.324 weighting methods equally and 0.329 weighting games equally, and its 95% bootstrap interval over games excludes zero in both cases ($[0.30, 0.35]$ and $[0.30, 0.36]$). Two qualifications must be stated alongside this result. When the oracle-equivalent intervention methods at faithfulness 1.0 are removed, leaving only coarser approximations, the gap shrinks but the causal and intervention family still leads, so part of the leaderboard's main contrast comes from methods close to the oracle operation itself. The gap is also regime-specific in a way that cuts against a simple "gradients fail" reading. On content outputs, where the naive gradient is valid, the gradient and correlational family is level with the causal and intervention family (0.642 against 0.637), so the content gap is -0.01, essentially zero. On the position regime, by contrast, the gap is 0.338 in the expected direction, and at 44 games its bootstrap interval $[0.29, 0.38]$ excludes zero. The conclusion is now two-sided: pooled over every output, and on the position regime alone, causal and intervention methods are more faithful than gradient and correlational methods, both with intervals that exclude zero. The position-regime advantage, once only directional, is now statistically established at the 44-game sample size. This

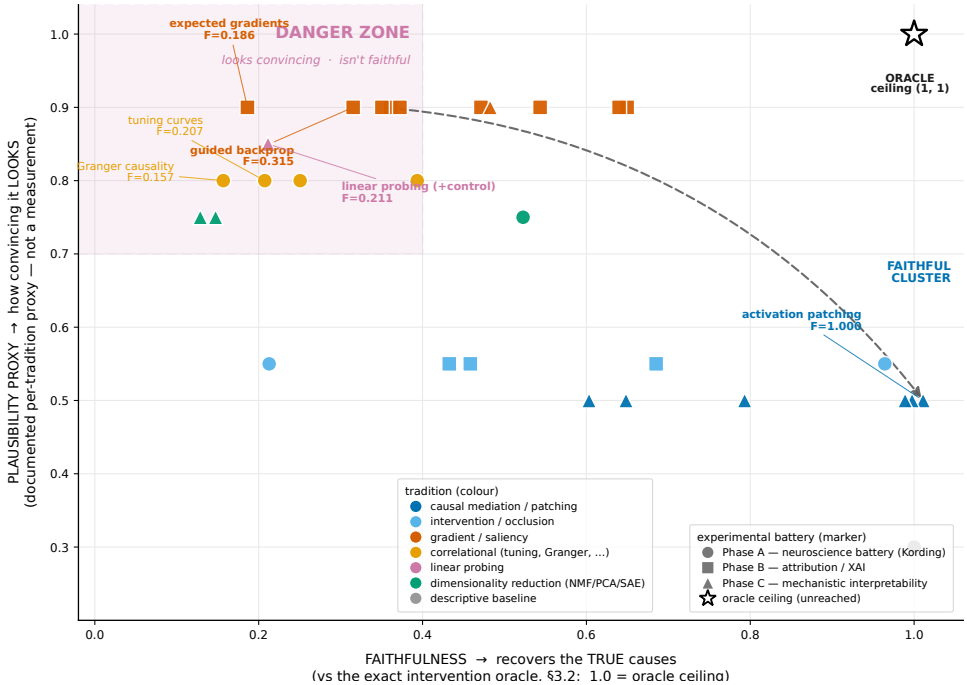


Fig. 2 Every method, every tradition, on the shared faithfulness-versus-plausibility plane. The plotted set is the canonical 30 interpretability methods + the oracle = 31 rows, each placed by its faithfulness against the intervention oracle (Section 3.2; X, exact ground truth) and by its documented method-tradition *plausibility proxy* (Y; Section 3). The plausibility axis is a tradition-level proxy, not a human-subjects measurement. The oracle positive control anchors (1.0, 1.0), a ceiling no interpretability method reaches. Pooled over all outputs the causal and intervention family is more faithful than the gradient and correlational family (0.71 against 0.38, a gap of 0.32 whose 95% interval excludes zero). Note that several gradient and correlational methods sit low on faithfulness yet *higher* on the proxy — the *danger zone* of high apparent plausibility at low faithfulness. Among the gradient methods the danger-zone offenders are expected gradients ($F = 0.19$) and guided backprop ($F = 0.32$) at plausibility 0.9; vanilla saliency is less extreme, since its all-regime faithfulness is 0.36. Every plotted value reads from the committed leaderboard, none re-run or estimated; per-method values with 95% confidence intervals are in Tables 1–3 and the supplement.

is the cross-tradition analogue of what neural-network benchmarks report when faithfulness is scored against a constructed ground truth [44, 45], with the difference that here the ground truth is an exact intervention on a real software artifact rather than a synthetic or pseudo-label.

The danger zone

If the faithfulness axis matched how convincing a method looks, it would tell us nothing new. It does not match. Plotted against the plausibility proxy of Section 3, the points split into a safe diagonal and a dangerous corner. The methods the field uses first sit low on faithfulness while having the highest plausibility-proxy values. Linear probing

sits in the danger zone at plausibility 0.85 against faithfulness 0.21; expected gradients follows at 0.9 against 0.186; A3 tuning at 0.8 against 0.207. This inversion is the key finding: a map that is weak or empty on the position regime looks *more* convincing than the exactly correct one. We state the caveat plainly. The plausibility axis is a documented method-tradition proxy assigned per tradition before the faithfulness scores were read, not a human-subjects measurement, and we report it as such to avoid overclaiming the danger zone. Still, the faithfulness axis is exact, and the ranking it inverts is what the field tends to trust, so the hazard is real even at the proxy's coarse resolution. The danger zone is a quantitative result, not a speculation.

Making the unfaithful faithful

A reader could accept the danger zone and still dismiss it by blaming the gradient. Under the naive read the saliency methods score zero on the position regime because the hard forward map reads a sprite's position through an integer index, so the gradient is zero almost everywhere (Section 1.2, Prop. 1). On this reading, the danger zone is a technical artifact a smoother forward pass would repair. Once repaired, the gradient methods would become faithful and the concern would disappear. The differentiable port lets us run exactly that repair and test the claim. We turn its bilinear sampler on, replacing the integer index read by a triangular-kernel occupancy whose sprite column is a continuous function of the position byte. This restores a non-zero gradient through the position computation without altering the forward output. It is the same sampler with which the battery scores the gradient rows' position outputs in Table 2.

The repair restores a non-zero gradient, but its faithfulness stays low and mixed (Fig. 3). On Pong, vanilla saliency, gradient $\times$ input, SmoothGrad, and expected gradients all rise from the provable zero to faithfulness 0.791 against the oracle; on Q*bert vanilla saliency, gradient $\times$ input, SmoothGrad, and integrated gradients rise to 0.529 (committed sampler record). The rise is the exception, not the rule. Averaged over the six position games the restored faithfulness is only about 0.22 for the SmoothGrad-family methods, 0.13 for expected gradients, and 0.09 for integrated gradients, because the gradient becomes non-zero on just one or two games. Integrated gradients with a zeros baseline still vanishes on Pong, the static-sprite games offer no position to restore, and on Space Invaders the restored gradient lands on a non-active cell, so it does not move. So turning the sampler on does not make the gradient methods faithful. It only removes the objection that their position score is a pure artifact of a non-differentiable read. Differentiating the whole horizon end to end does not help either (Section 1.2). A faithful-capable substrate is not the same as a faithful method.

But this does not give understanding. The restored gradient on Pong concentrates entirely on the single RAM byte that holds the paddle's position, `ram[54]`, which the oracle independently confirms is a true cause. The method now finds the right cause and attributes nothing to the wrong ones. Yet it still recovers no meaning, exactly as when the map was empty, and exactly as no other method in the study recovers any (same committed record). A faithful attribution to `ram[54]` names a byte the program reads. It does not say that the byte *is* the paddle, that reading it *is* tracking, or that the routine computes a bounce. The only semantic label in play is the imported annotation we use to *check* where the attribution landed, never one the method *produced*. Even

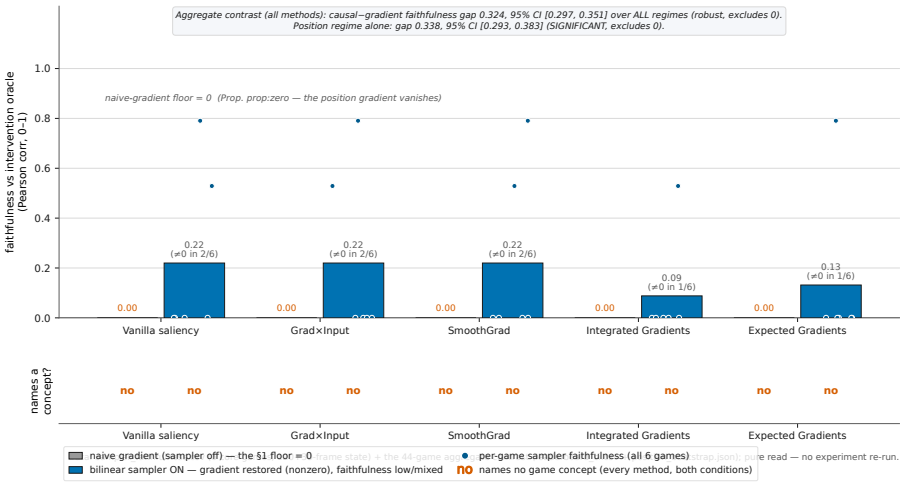


Fig. 3 A differentiable substrate restores a non-zero gradient, but the restored faithfulness stays low and mixed, and no method names a game concept. Top: on the position regime the naive gradient is provably zero (Prop. 1), so the saliency methods reach faithfulness 0 (naive bars); turning the differentiable port's bilinear sampler on restores the gradient through the position read and lifts faithfulness to 0.791 on Pong and 0.529 on Q*bert for the methods with a moving tracked sprite (sampler bars), the restored gradient concentrating on the true position byte (`ram[54]` on Pong, oracle-confirmed). Note that the rise is confined to one or two games: averaged over the six position games the restored faithfulness is about 0.22 for the SmoothGrad-family methods 0.13 for expected gradients, and 0.09 for integrated gradients, so a faithful-capable substrate does not make the gradient method faithful. **Bottom:** semantic recovery is a yes/no question, and the answer is no in both conditions and for every method: an attribution map names a cause but emits no game concept; the only semantic labels are imported (Section 3.2) and used to check localization, never produced. The rise is reported only where a tracked sprite exists: integrated gradients with a zeros baseline vanishes on Pong, static-sprite games have no position to restore, and Space Invaders' restored gradient lands on a non-active cell. Values read from the committed sampler record; none re-run.

where the restored gradient is faithful, it carries no meaning. This result sets up the rest of the section: the gap we measure next is not the unfaithful methods' deficit, because restoring the gradient does not close it (Fig. 3).

Where the leaderboard stops — the semantic gap

Here the paper turns. Take the best case for the audit: a method that scores 1.0 against the oracle. We ask whether it gives an understanding of what the program does. It has not, and we are careful about what that claim is and is not. We do *not* claim that no method can recover meaning; that is the unbounded statement, the subject of the companion design-recovery work, which we have not run here. We claim the bounded thing the oracle can prove: even when a method recovers the true causal structure exactly, what it returns is the causal structure — a causal-effect map, a data-flow graph — and not the *semantics*, the variable's meaning or the routine's algorithm. The missing object is a different kind of thing. It is not just a poorer version of the causal

structure. Following the triad of Section 3.4, we operationalize the gap through three separations, each adjudicated by the oracle: faithful-not-sufficient, named-not-causal, and decodable-not-used. They diagnose distinct semantic failure modes; they are not a complete theory of semantics. The three separations show the failures pointing in opposite directions (Fig. 4).

The first separation is graph-correct yet behavior-incomplete: high F and M, low S. Automatic circuit discovery (ACDC) [20] on Q*bert recovers a data-flow graph that is, by the exact data-flow oracle, perfectly correct and perfectly parsimonious, $F = 1.0$ and $M = 1.0$ [35]. Yet the discovered circuit alone, with its non-circuit parents scrubbed, reproduces the clean candidate-cell readouts only 10% of the time, $S = 0.10$. A correct graph of which cells feed which other cells is not the function those cells compute. The faithful side shows the same. Activation patching [17] recovers the exact per-site causal-effect table on every one of the core games [35], and a table of effects, however exact, is not the bounce algorithm that produced them. No tuning budget is left to spend here, because the effect table is *already* exact, so the residual to the algorithm cannot be a worse fit closed by a better hyperparameter. It is a missing object of a different type, and that is why the gap is not merely "methods underperform."

The second separation is named yet causally unused, the reverse failure: high matched fraction, low causal F. A sparse autoencoder (SAE) [23] trained on the recorded Pong state matches all four candidate game-concept variables that vary on Pong, a feature-to-variable matched fraction of 1.0, a perfect semantic alignment by the field's usual yardstick. Pong sits at the top of the range: over the 42 SAE games the matched fraction ranges from 0.5 to 1.0 (mean 0.88), and 20 games reach 1.0. By the oracle, those same matched features are nearly causally unused on Pong: faithfulness $F = 0.041$ and a *negative* sufficiency $S = -0.46$ [35]. A feature that looks like the ball's x is not the same as the mechanism the program uses to compute with the ball's x. The circuit was correct but incomplete; the dictionary is named but causally wrong. The gap runs in the opposite direction on the same axis. An account of methods simply scoring too low cannot explain this. Recent SAE evaluations report the same proxy-versus-use gap on neural language models [29]; the oracle here turns that observation into an exact causal-use test on a known system. We guard against reading semantic success into an aligned variable. The interchange-intervention alignment (DAS) [22] reports an aligned-variable accuracy of 1.0, but that 1.0 is true *by construction*, verifying a *supplied* variable rather than discovering one. A method that confirms a given concept is not a method that found the meaning.

The third separation is decodable yet unused, the present-versus-used trap, where the oracle is strictly stronger than the probe: high probe accuracy and selectivity, low intervention effect. Linear probing with control tasks [25] decodes game concepts from the Breakout state at a mean accuracy of 0.96 with positive selectivity over its control; the information is demonstrably *present*. Yet the exact oracle flags a score-related cell that is *present but unused* (cell 84): re-derived within the horizon, its value has no downstream effect on the output [35]. A concept can be linearly readable yet play no part in the computation, exactly Hewitt and Liang's control-task warning; decodability is necessary, never sufficient. Probing answers whether the concept is present. The

oracle answers whether it is used. Only the second tells us how the program works. Linear probing's leaderboard faithfulness of 0.21 paired with a plausibility proxy of 0.85 is its place in the danger zone, and this cell is the mechanism behind that placement.

Read together, the leaderboard plane of Fig. 2 and the failure taxonomy of Fig. 4 make the same point from three sides, and the triad's axes are not redundant: a method can be faithful and minimal yet not sufficient (ACDC on Q*bert, $F = M = 1.0$, $S = 0.10$); it can match a named concept perfectly yet be causally unused (the SAE on Pong, matched fraction 1.0, $F = 0.041$); it can flag a concept the program never uses (cell 84, present but unused). The claim does not rest on any single low number; a lone low S could be a tuning artifact. It rests on two things. First, the gap appears where tuning cannot help: the effect table is already exact. Second, the failures run in opposite directions on the same axis. A "methods are not good enough yet" account cannot explain this two-sided result. We restate the honesty contract here, because this is the easiest place to overread the results. We have *not* recovered the program's meaning, algorithm, variables, or design from scratch. Every semantic label we used is external, imported from AtariARI [32] and OCAtari [33]. The substrate only *causally verifies* a supplied label, it does not *discover* the concept, and the provenance record marks each label as imported rather than recovered. So the faithful methods recover the causal structure. The meaning is a separate thing, and we have measured the distance to it. Recovering that meaning is a different problem, and it is the companion paper's, not this one's.

A note on aggregation closes the section, because two committed records report the sparse autoencoder differently and a careful reader will see both. The leaderboard score [35] aggregates over the per-method records, which for the SAE gives faithfulness 0.147 with a 95% bootstrap interval of $[0.09, 0.21]$ that brackets it; the figures plot this value (Supplement S5 records both aggregations). The per-game Pong record used in the named-not-causal separation above is a different, finer object, $F = 0.041$ on that one game. We keep the two distinct: the 0.147 is the leaderboard aggregate, the 0.041 is the single-game illustration. Throughout the paper, the leaderboard and its figures use the per-method aggregation as canonical, so text, table, and figure agree.

2 Discussion

We applied the interpretability toolkit to a machine whose ground truth is computable. This let us measure two things at once. On this system the true cause of any output is recoverable by exact intervention. The causal and intervention methods are faithful, while the field's familiar gradient and correlational tools are not. The leaderboard separates them by faithfulness, the one axis we can score exactly (Fig. 2). Pooled over all outputs the causal and intervention methods reach 0.71 against the gradient and correlational methods' 0.38, a gap of 0.32 whose 95% interval excludes zero. On the discrete position outputs a causal method reaches 1.000 where standard saliency reaches only 0.092, on a machine whose answer is known exactly. Yet faithfulness is the easier of the two measurements. Even a method at the oracle ceiling returns only the causal wiring: an effect map and a data-flow graph. It does not return the semantics:

the meaning of a variable or the algorithm of a routine. We call this residual the *semantic gap*, and it is a number we can measure.

We score three traditions with one common oracle: the neuroscience analyses, the attribution toolkit, and mechanistic interpretability. These traditions are rarely studied together, but the oracle places them on one scale. Jonas and Kording warned that an analyst armed with systems neuroscience could fail to understand even a microprocessor, because describing structure is not the same as explaining how it works [4]. We make that a measurement. The classical analyses recover rich structure yet score low against the known mechanism. The naive gradient is provably zero on the discrete position regime, and only the causal operators reach the ceiling. Mechanistic interpretability has until now been calibrated against synthetic or compiled ground truth, transformers with a planted circuit by construction [53, 54], ground-truth attribution benchmarks that plant the important features [45, 55], and circuit- and variable-localization benchmarks that curate their answers in neural language models [44]. We add the missing case: a real, human-written, deployed artifact that was never designed to be interpretable. Its ground truth is independent of the artifact, not built into it. Our subject is not the Atari environment itself, which the Arcade Learning Environment and its faster and object-centric descendants already supply [56–59]; it is the bit-exact, differentiable intervention oracle layered on it. So the leaderboard calibrates these methods against ground truth on a system the field did not design to be interpretable.

The same oracle also reveals a limit: a high leaderboard score cannot show meaning. This limit is the subject of this paper. A perfect faithfulness score certifies that a method named the true causes. It does not certify that the method recovered what the program means. We observed this limit in three cases (Fig. 4). On Q*bert a circuit is graph-correct and minimal ($F = M = 1.0$) yet reproduces behavior only 10% of the time ($S = 0.10$). On Pong a feature dictionary matches every named variable (matched fraction 1.0) while being nearly causally unused ($F = 0.041$, $S = -0.46$). And a concept is linearly decodable yet, by the oracle, never used in the computation. The gap appears in two opposite forms: correct-but-incomplete, and named-but-wrong. A simple "the methods are not good enough yet" story cannot produce both.

We tested the strongest version of that story and it failed. The faithfulness gap could be read as an artifact of one design choice. The discrete position read-out makes the naive gradient vanish by Prop. 1. So the gradient family scores a provable zero on the position regime, and the failure could be dismissed as a fixable technical artifact. We removed that artifact. Turning the port's bilinear sampler on restores a usable gradient through the position index. On the games with a moving tracked sprite the restored gradient becomes faithful. On Pong it reaches 0.791 for vanilla saliency, grad×input, smoothgrad, and expected gradients. On Q*bert it reaches 0.529 for vanilla saliency, grad×input, smoothgrad, and integrated gradients (Fig. 3). The restored gradient concentrates on the position byte that drives the sprite, `ram[54]` on Pong; it finds the right cause. Yet the rise is confined to one or two games. Averaged over the six position games the restored faithfulness is only about 0.22 for the SmoothGrad-family methods, 0.09 for integrated gradients, and 0.13 for expected gradients. So a faithful-capable substrate does not make the gradient method faithful in

general. Where it does become faithful, none of these methods recovers any meaning, exactly as before, and exactly like every other method in the study. Even where the gradient is corrected, it identifies the cause but no meaning. So the meaning was never present in the wiring for a method to recover.

A second artifact reading also fails. The gradient methods differentiate a single read at the intervention frame, so one could argue that a gradient through the running program would do better. We computed that gradient on all 44 scored games, through up to 10^5 bit-exact CPU instructions per game (Section 1.2). On most games it returned the same map as the single read and the same scores. On 8 games it lost the content gradient and on 8 the position gradient entirely. It helped once, on Centipede, where the output byte is carried over from the oracle's largest cause. The program's causal structure lives in branches, indexed addressing, and hardware reads, and backpropagation does not traverse any of them. A deeper gradient is not a more faithful one.

The improvement occurs only in specific cases, and that pattern is itself evidence. It is not a universal rise. It rises only on the games with a moving tracked sprite the gradient can flow to. Integrated gradients with a zeros baseline still vanishes on Pong. The static-sprite games show no rise, and so does the case where the restored gradient falls on a cell the program does not currently read. Where a position byte exists and moves, the sampler makes that gradient faithful. We gave the faithfulness explanation every case it could win. None of it produced a single semantic label (Fig. 3).

What understanding would require

Faithfulness stops short for a reason that Marr's levels of analysis make clear [60]. A faithful attribution lives at the implementational level, where cells and registers carry the signal. Understanding lives one level up, at the algorithmic level, in what function the program computes. A correct data-flow graph is an implementational fact. The bounce algorithm it implements is an algorithmic fact, and the two are objects of a different type. Lazebnik made the same point: one can describe a system in complete mechanistic detail and still not understand it [61]. We turn that observation into a measured quantity. The distance from a faithful effect table to the algorithm is the sufficiency gap $1 - S$, and on an already-exact effect table it cannot be closed by a better hyperparameter. Our operational definition of interpretability, ground-truth recovery scored by the oracle, is one of several in circulation; others set it as a design target for new models rather than an audit of a system already built [62].

Software engineering gives the clearest version of this requirement. IEEE 610.12 defines software as programs plus their documentation and data, the specification, the design, the intent [34]. A faithful circuit recovers part of the program. To understand the system you must also recover its documentation: what each variable is for and what each routine should do. The sampler-on result tells us why no internal method can reach that object. The meaning of a variable is not a causal fact about the variable. It is a fact about the behavior the design serves, and that behavior is fixed outside the program code, in the manuals and the data the program was written against. A method that reads only the mechanism, however faithful, reads only causes. A cause becomes a meaning only against an external reference: the system's documented behavior. No

perturbation of the chip can reach that reference. In this study the only semantics present entered from outside: every T3 label was imported from AtariARI and OCAtari and then merely verified to be causally live, never discovered by a method. Hence the gap is not an artifact of weak methods but a property of the question. Faithfulness is necessary for understanding but not sufficient, because understanding needs an external behavioral reference that an internal method does not carry. We state clearly that this paper defines and measures the gap. Closing the gap means recovering the program's documentation and design from the bare machine. That is the reverse-engineering goal, and the task of the companion design-recovery paper (Paper 4). That paper measures the distance to that goal rather than clearing it. Whether a recovered concept matches how the system is used is the companion behavior paper (Paper 3). The learned agent, where there is no disassembly to check against, is Paper 5. Papers 3–5 are unpublished companion works, and no claim here relies on them.

Probing reads the same result from the data side. It shows that a concept is present in the state, but not that the concept is used. This is the control-task warning of Hewitt and Liang [25]. Our intervention oracle is strictly stronger, because it asks whether the byte causes the output, and it found a present-but-unused score cell the program never reads. Decodability is presence; use is an algorithmic fact. Understanding needs use, and only the oracle supplies it. At base, the three traditions are one toolkit under different names. Tuning curves match feature visualization, lesions match ablations, and Granger causality is the correlational predecessor of causal attribution. Each pair shares a failure mode, so a result measured on one transfers as a caution to the others. A benchmark built on an Atari chip thus speaks to systems neuroscience, explainable AI, mechanistic interpretability, and software verification at once.

Why a 1977 chip is a fair screen

We address the obvious objection. The VCS is deterministic, hand-coded, and small, while the systems these methods target are learned, distributed, stochastic, and large. We do not ask a score here to predict a transformer. The benchmark is a necessary-condition stress test, not a sufficiency predictor. Here a method has perfect ground truth, full observability, exact interventions, and no learning to complicate the result. If it still fails, that failure is strong evidence against any claim that the same method reliably recovers causal mechanisms in harder, less observable systems. Passing here is necessary but not sufficient; failing here is informative. The same logic suggests, rather than proves, a reading of the semantic gap. If even the strongest causal account a fully observable machine permits stops short of meaning, the result suggests that faithfulness alone should not be treated as understanding in less observable neural systems, where the circuit itself is uncertain and no clean ground truth is on hand. We measure $1 - S = 0.90$ on an exactly recovered circuit on Q*bert ($1 - 0.10$; the ACDC family mean is $1 - 0.31 = 0.69$), and we do not extend that number to neural networks as a proven floor; the precise scope, and what this paper does not claim, are set out in the Supplementary box on what we do and do not measure (S6).

The chip also exhibits the properties that make real systems hard to interpret. There are aliased, polysemantic RAM cells reused across routines, a hand-coded analogue of superposition. There is computation by racing the electron beam, which is

distributed and temporal. And there are registers that mean different things in different contexts. Figure 5 maps each VCS property to the neural-network difficulty it does and does not capture, so that a property we can prove on the VCS is one to suspect in neural networks. We state plainly what the chip does not capture: learned features, scale, and stochasticity are absent. The learned agent that adds all three back is Paper 5.

What interpretability needs

Three recommendations follow. The field should prefer causal over correlational evidence, because the methods it reaches for first carry the highest plausibility proxy and the lowest measured faithfulness, and a saliency map provably empty on the position regime can look more convincing than the exactly correct map. It should validate on known systems before trusting on unknown ones, since a method that cannot pass where the answer is computable cannot be audited where it is not. Most important, it should stop treating faithfulness as understanding. The open problem for explainable AI is not a better saliency map; it is the semantic gap. We confirmed this directly: where we closed the faithfulness gap for the gradient methods, the meaning still did not appear. The field does not need an instrument that reads the mechanism more sharply. It needs one that supplies the reference the mechanism does not contain: the system's behavior, documented for software in its manuals and data. Faithfulness is the part we can now measure, and on causal methods we can achieve it. Meaning remains unsolved, because no internal method carries the external reference it needs.

We have measured a gap, not closed it, and several limits bound what we may conclude. Our claim is therefore the bounded one: on exact ground truth, faithfulness certifies the causal structure, and that structure alone is not the semantics. We do not make the unbounded claim that "no method can recover meaning," which is Paper 4's charge. The sufficiency and minimality axes are defined against the oracle, and the T3 semantic labels are external, imported from AtariARI [32] and OCAtari [33] and only causally verified, never discovered. A single low sufficiency could be a tuning artifact. So the claim does not rest on any single number. It rests on three facts: the gap appears where tuning cannot reach it; it runs in both directions; and it survives the sampler-on test that makes the gradient methods faithful on games with a moving tracked sprite. We do not read that test as universal. The rise occurs on Pong and Q*bert, but not on the static-sprite games, not on a zeros-baseline integrated gradient, and not where the restored gradient falls on a cell the program does not read. We report it as such, not as a law over all methods and games. The plausibility axis is a documented method-tradition proxy, not a human-subjects measurement, and we report it at that proxy's coarse resolution. Paper 1 validated the port against `xitari` over 30 RAM frames and 60 screen frames. Our shared testbed scores at frame 105 (a 90-frame gameplay prefix plus the 15-frame horizon), beyond that window, so we validated the analysed streams themselves: on all 44 scored games the port matches `xitari` on every frame up to 105, in RAM and in screen. Every scored intervention is also re-run bit-exactly on the port itself. The subject is one old machine with no learning, which is the necessary-condition framing once more: a stress test, not a forecast.

We end with the broader implication. Because the substrate is differentiable, it tells us which gradient explanations are trustworthy and where they vanish. The content path carries an honest gradient while the discrete index does not. So a saliency map over sprite position is empty for a structural reason, not by chance. This statement about gradients generalizes to any system where a discrete decision follows a continuous one. We believe a fully known, differentiable testbed is valuable for exactly this reason: it lets the community calibrate its instruments before deploying them where calibration is impossible. The hard case remains the learned agent, where there is no disassembly, no specification, and no engineer's intent to recover, and where the semantic gap is likely to be larger. Closing this gap may take the next decade of interpretability research. The goal is to turn a faithful account of a learned system into an understanding of what it has learned. We have not solved this problem. We have defined and measured it, which is the necessary first step.

3 Methods

Our method rests on a single idea. We study an interpretability question on a machine whose answer we can compute exactly. We describe in turn the substrate that makes the ground truth obtainable, the oracle that delivers it, the tiered semantics we attach to it, and the correctness triad that scores understanding as numbers. The per-method setups live in the Supplementary Information. Fig. 1 sketches the pipeline.

3.1 The differentiable, bit-exact VCS substrate

The subject of this study is the Atari Video Computer System (VCS) itself, not a learned agent. It is a real, complex artifact never designed to be interpretable. Yet we can measure its every causal dependency exactly. Paper 1 built it, a bit-exact, differentiable port of the 1977 VCS. The port is validated bit-for-bit against the `xitari` C++ reference on all 64 games of the Paper-1 roster, in random-access memory and rendered screen, over a 30-frame RAM and 60-frame screen conformance window [30, 31]. The port exists in two cross-checked realizations, `jutari` (Julia) and `jaxtari` (JAX/Python). All scored runs in this paper use `jutari`.

The substrate grants four capabilities that no neural network offers for free. The first is an exact forward map,

$$y \;=\; \mathrm{VCS}\big(\mathrm{ROM},\, \mathrm{inputs}[0{:}t],\, s_0\big), \tag{1}$$

that is, the output y at step t, a pixel, a score, or a game event, is a deterministic function of the cartridge bytes, the inputs up to t, and the initial state s_0. Re-running reproduces y to the bit. The second is exact interventions. We set any cause u, a ROM byte, a RAM cell, a CPU/TIA/RIOT register, an opcode (a CPU instruction byte), or a joystick input, to any value and re-run Eq. (1) deterministically. This realizes the causal operator $\mathrm{do}(u)$ on a real machine rather than an assumed world model. The third is full state observability. We record every bit of RAM and every register along the trajectory, so the "activations" we interpret are the program's own variables. The fourth is a content-path gradient $\partial y/\partial u$ through a straight-through

estimator (STE, a surrogate gradient whose forward pass still returns the exact discrete value) formulation whose forward pass is bit-exact to the hard machine at any finite temperature (a smoothing parameter) [63, 64], computed by reverse-mode automatic differentiation (Zygote) on the CPU. A neural interpretability study works with three things: the inputs, the state, and the outputs. Our substrate gives us all three.

We close one gap explicitly. Paper 1 certifies bit-exactness over 30 frames of RAM and 60 frames of screen on fixed action streams. Our shared testbed scores at frame 105, a 90-frame seeded random-action prefix plus a 15-frame horizon, which lies beyond that window. We therefore validated the exact streams we analyse. For each of the 44 scored games we replay its testbed stream through the same boot and step code the methods use, and compare RAM and screen against the seed-patched `xitari` reference on every frame from boot to frame 105. All 44 games match on every frame, in RAM and in screen (Supplement S3.2). Every game boots with its own ROM settings, including the start-up inputs the Arcade Learning Environment prescribes for some games, the same settings Paper 1's conformance suite uses. Within the port, the oracle also re-runs each scored state twice from boot and asserts that RAM and screen are bit-identical before it trusts any effect. The gradient methods differentiate the output read at the intervention frame. As a control, we also differentiate the output at the end of the 15-frame horizon back through every CPU instruction in between (Section 1.2). We do not assume gradients over longer horizons.

3.2 The ground-truth oracle

The oracle is central to the method. For any output y and cause u, it returns the true effect of u on y. We score every method against it. Its primary construction is the exact intervention oracle: we intervene on u by occlusion, clamping, replacement, or resampling, re-run the machine deterministically, and record the change in the output,

$$\Delta y(u) \;=\; \mathrm{VCS}\big(\ldots,\, \mathrm{do}(u{:=}v'),\, \ldots\big) \;-\; y. \tag{2}$$

The true causal effect of a cause is the difference its edit makes to the output. We measure it by running the altered machine. We assume no world model and no smoothness. The collection $\{\Delta y(u)\}$ over all causes is the true causal map of y, the reference for the faithfulness axis of every method.

The output y comes in five regimes. The regime fixes what a method must predict and whether a gradient is even meaningful. A *content* output is a register, color, or graphics-bit value that flows smoothly into the pixel, where the naive gradient is valid. A *position/index* output is a sprite, missile, or ball coordinate set by strobe timing. Here the naive gradient vanishes by Prop. 1, and only the bilinear-sampler surrogate or the intervention oracle applies. An *event* output is a discrete state change, a collision latch (a one-bit hardware flag) or a life lost, also gradient-zero. A *score* output is the game's running counter, valid for the gradient only where it varies continuously. A *pixel* output is the rendered framebuffer (the array of screen pixels) value. It is content-valid where it tracks a smooth cause and index-zero where it tracks a position. The Supplement lists, per game, which outputs each method is scored on and the per-output gradient verdict.

The oracle uses several kinds of intervention. We list each one, because the kind limits what a score means. Occlusion and clamping are surgical but *off-manifold*: they can set a byte to a value the running program would never write. Replacement from a donor trajectory is *on-manifold*: the substituted value is one the machine actually produced elsewhere. Resampling draws the cause from its own marginal over the trajectory. Counterfactual edits are valid under the emulator by construction, since any byte assignment yields a well-defined deterministic re-run. Off-manifold interventions are still valid for oracle scoring. The oracle measures a causal effect, not a likelihood. We ask what the output would be if u were v'. The bit-exact machine answers this exactly for any v', on- or off-manifold. Where a method's own claim is about typical behavior rather than causal dependence we fall back to the on-manifold donor and resampling variants and flag the choice in the Supplement.

The content-path gradient $\partial y/\partial u$ is the second way to build the oracle. We read it through the differentiable substrate. It is fast and dense, and it agrees with the intervention oracle on the content path, the register, color, and graphics-bit values that flow smoothly into the pixel. Yet, one important limit applies to the whole study. The straight-through gradient routes to the content path only. So a discrete index, a sprite's screen position the VCS sets by the timing of a strobe write (a write that latches the sprite's horizontal position), has a naive gradient that vanishes under the hard forward map. We make this precise.

Proposition 1 (Zero naive gradient for discrete-index outputs)**.** *Let an output y depend on a cause u only through a discrete index $n(u) = \lfloor g(u) \rfloor \in \mathbb{Z}$, the integer count of color clocks (each one pixel-time of the scanning beam) between a strobe write and the position-counter reset that the VCS uses to place a sprite, missile, or ball, with g continuous, piecewise linear, and non-constant (strictly monotone on each linear piece). Under the hard (bit-exact) forward map the rendered output is $y = h(n(u))$ for a fixed render function h. Then $n(\cdot)$ is locally constant wherever $g(u) \notin \mathbb{Z}$, so $\partial y/\partial u = 0$ at every such point, and y is discontinuous on the set $\{u : g(u) \in \mathbb{Z}\}$, which is measure zero precisely because g is strictly monotone on each piece, and there the gradient is undefined. Hence the naive gradient is zero wherever the derivative exists: a tiny change in u cannot move the sprite by part of a pixel. Any method that reduces to this naive gradient inherits the failure on position, index, and event outputs.*

The proposition states only the hard-map limit; it does not say that no useful gradient exists. Two surrogate constructions of Paper 1 [30] are explicit exceptions and are *not* covered by it: the straight-through estimator, whose forward pass stays bit-exact to the hard machine at any finite temperature while routing a non-zero gradient along the content path [63, 64], and the bilinear sampler that interpolates the index boundary to recover a usable position gradient [65]. The companion paper constructs both and validates the bit-exactness this proposition assumes. We import that construction and prove the vanishing inline, so the present paper stands alone. We therefore avoid any blanket claim that "gradients are provably zero." We use instead the scoped claim of Prop. 1: the *naive* gradient for discrete position/index outputs is zero under the hard forward map, and methods that reduce to it inherit that failure. Surrogate and bilinear-sampler gradients are exceptions.

We make the faithfulness audit as strict as possible. The position regime leaves one objection open, and we close it. A reader might object that the naive zero of Prop. 1 is just a fixable artifact. So we fix it: we turn on the Paper-1 bilinear sampler. The sampler interpolates the index boundary of Prop. 1 in the spatial-transformer manner. This makes the discrete position output a continuous function of the position RAM byte rather than a step. So the gradient methods receive a non-zero surrogate position gradient where the hard map gave them nothing [65]. We score both conditions, the naive forward and the sampler-on forward, against the same intervention oracle of Eq. (2). For each candidate cause the per-cause attribution is correlated, by the raw Pearson ρ of Section 3.4, against the oracle's true $|\Delta$ position-pixel$|$. So the two conditions differ only in the surrogate the method is handed, not in how it is judged. Alongside faithfulness we record a second, deliberately stricter property. The *semantic recovery* of a method is a yes/no question: does the method, on its own, emit a game-concept name, the word "ball" or "missile" attached to the byte it points at, rather than a cause that an external label only names later? An attribution map returns weights over causes and never a name, so for it the answer is no by construction: it recovers no meaning. Faithfulness and semantic recovery are independent, and they are of different kinds. Faithfulness is a graded score for whether the method names the right cause. Semantic recovery is a yes/no property: whether it names any meaning at all. The keystone experiment of Section 1.4 reports both for the naive and sampler-on conditions on the position regime.

Hence, for content outputs the gradient is a legitimate companion oracle, while for position, index, and event outputs the intervention oracle of Eq. (2) is the sole ground truth and every gradient-based method is scored as a method under test. This position regime is where the sharpest separations in our results appear. Where the two overlap we report their correlation and flag, rather than hide, the points at which they disagree.

3.3 Tiered ground truth and how we obtain it

The ground truth comes in three tiers. The tiers are not just bookkeeping; they support the paper's central argument. Tier 1 (T1) is the exact causal map of Eq. (2), derived internally from the substrate and available for all 64 games. Tier 2 (T2) is the hardware roles and the read/write data-flow graph, also exact and internal across all 64 games. T1 and T2 describe what the machine does internally, and need no external label.

Tier 3 (T3) is different in kind. This difference matters for our argument. It attaches a game-concept name to a cell or feature, "the ball's x position" or "the score." These names cannot be computed from the substrate alone. They are facts about what the program *means*, which is separate from what it *computes*. We import T3 from two public sources and make it causal. AtariARI provides labels for 22 games, read from commented disassemblies, with raw RAM coordinates not aligned to the rendered position [32]; OCAtari covers more than 40 games and, in its RAM mode, applies the render-time offsets that fix that misalignment [33]. We import both as candidates and verify each causally by setting the byte, re-rendering, and confirming on the bit-exact framebuffer that the named object moves to the predicted place. This upgrades a correlational label to a verified-causal one and auto-corrects the offsets; further

labels we discover by RAM-to-framebuffer correlation and intervention. The headline core set is six games, Pong, Breakout, Space Invaders, Seaquest, Ms. Pac-Man, and Q*bert, each carrying both sources, all in the Paper-1 bit-exact roster. Together they span paddle-and-ball, shooter-with-shields, resource-meter, multi-agent-pursuit, and discrete-grid logic. The remaining games form a breadth set scored on T1 and T2 only.

We are careful about what this gives us. All T3 labels are external and only verified, never discovered from scratch. So a method that matches a supplied concept has aligned to a given meaning, not recovered it. This is exactly why a probe can show that information is *present* without showing it is *used* [25]. Our intervention test asks whether the byte *causes* the named object to move, so it is strictly stronger than decoding it. Recovering the program's documentation and design from the bare machine is a separate problem, the subject of the companion paper.

3.4 The F ∧ S ∧ M correctness triad

A faithful attribution is not yet an understanding. To measure understanding, we need an operational criterion. We adopt one in the spirit of Marr's levels of analysis [60] and of Jonas and Kording's warning that finding structure is not the same as explaining it [4]: each axis is first a continuous score, and the boolean "right" is a thresholded reading of those scores. We score an explanation $\hat{E}$ of how an output y arises by three numbers, faithfulness $\mathrm{F}_{\mathrm{score}}(\hat{E})$ and minimality $\mathrm{M}_{\mathrm{score}}(\hat{E})$ in $[0,1]$ and the held-out sufficiency $\mathrm{S}_{\mathrm{score}}(\hat{E})$ in $[-1,1]$, and call it *right* when all three clear their thresholds,

$$\mathrm{right}(\hat{E}) \;=\; \mathbf{1}[\,\mathrm{F}_{\mathrm{score}} \geq \tau_{\mathrm{F}} \;\wedge\; \mathrm{S}_{\mathrm{score}} \geq \tau_{\mathrm{S}} \;\wedge\; \mathrm{M}_{\mathrm{score}} \geq \tau_{\mathrm{M}}\,], \tag{3}$$

with $\tau_{\mathrm{F}} = \tau_{\mathrm{S}} = \tau_{\mathrm{M}} = 0.5$ throughout. Results report the continuous scores; the predicate is a derived summary, not the primary quantity. The leaderboard's composite $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ column is the geometric mean of a method's three mean scores (the plain mean of those present when a leg is missing). All three are defined against the oracle, not against any method under test, which keeps the criterion non-circular.

Faithfulness asks whether the claimed causes are the *true* causes, scored against the intervention oracle of Eq. (2). The per-record metric depends on the explanation's output type, so $\mathrm{F}_{\mathrm{score}}$ is a single number assembled from three type-matched primitives. The committed score is the bare effect-agreement primitive, not a rescaled one. For a real-valued attribution map we report the Pearson (or Spearman, for ranked outputs) correlation $\rho(\hat{E}, \{\Delta y(u)\})$ between the attribution and the true causal effects, scored *raw*. A method whose attribution carries no causal signal therefore lands at $\rho = 0$, not at the 0.5 of an affine $\frac{1}{2}(1+\rho)$ remap. This is what makes the position regime informative. Under Prop. 1 the naive gradient is identically zero there, and its correlation with the oracle is 0. The reported faithfulness is therefore 0.00 and not a 0.5 floor. Each per-record file keeps the raw correlation, so a negative correlation is stored as a negative number. The leaderboard summarizes all three phases the same way: it clips each per-record correlation at 0 and then averages. Negative records occur in every phase, so the clip matters. Occlusion scores $F = 0.685$ clipped and 0.681 raw; the tuning analysis A3 scores 0.207 clipped and 0.117 raw. For the same attribution map we also report, as quality measures, deletion/insertion area-under-curve (deleting or inserting

the top-attributed inputs and tracking the change in output) and precision at k against the true causal top-k, $\mathrm{P@}k = |\,\mathrm{top}_k(\hat{E}) \cap \mathrm{top}_k(\{\Delta y(u)\})\,|/k$. For a graph-valued explanation (Phase A connectomics, Phase C circuit discovery) we use the edge F_1 against the true data-flow graph, the harmonic mean of precision and recall over recovered edges. For Phase C patching and mediation we use the causal-effect-agreement correlation against the true $\{\Delta y(u)\}$. The final $\mathrm{F}_{\mathrm{score}}$ is the per-record primary metric averaged first within a game over its output records, then with equal weight across the scored games, so no game-rich method dominates; confidence intervals are 95% bootstrap over games (Section 3.5). The three Phases differ only in which primitive is primary: Phase A uses graph F_1 for the data-flow methods and the correlation for the single-unit and tuning analyses, Phase B uses the effect-agreement correlation ρ with deletion/insertion AUC and P@k as the attribution map's quality measures, and Phase C uses the causal-effect-agreement correlation for patching and mediation and graph F_1 for circuit discovery. The leaderboard score is therefore a single scalar per method, but it is multi-metric in origin, and the per-method choice is recorded in the Supplement.

Sufficiency asks whether the explanation could regenerate the behavior under interventions it has not seen. We treat $\mathrm{S}_{\mathrm{score}}$ as a held-out predictive estimator: the explanation's per-cause claim, its attribution or the effect read off its causal sites, is read on a calibration split of interventions and used to predict the true effect on a disjoint held-out split of $\mathrm{do}(u)$ edits, each compared to the bit-exact re-run. We score it by the held-out predictive *correlation* between the claimed and the true Δy on that held-out split (Pearson, or Spearman for ranked outputs), so $\mathrm{S}_{\mathrm{score}} \in [-1, 1]$: a value near zero means the explanation predicts no better than the unperturbed output, and a negative value means it predicts *worse*, its held-out claim anti-correlated with the truth (the sparse autoencoder reaches $\mathrm{S} = -0.46$ on Pong, Section 1.3). Where the output admits only an exact match, a discrete index or event, we report instead the fraction of held-out interventions predicted correctly within a fixed tolerance, the $[0, 1]$ special case of the same held-out test. We deliberately do not define it through interchange interventions or scrubbing. Those are themselves mechanistic-interpretability methods under test in Phase C, so using them would make the criterion circular. All three phases report S where the calibration and held-out split is defined, and its bootstrap intervals follow Section 3.5. Like F, the leaderboard clips each per-record S at 0 before averaging, so a method-level S lies in $[0, 1]$.

Minimality asks whether the explanation is parsimonious and lives at the algorithmic level of the machine's own module decomposition. We define $\mathrm{M}_{\mathrm{score}} = |U^\star|/|U_{\hat{E}}| \in (0, 1]$, the ratio of the size of the true minimal cause set $U^\star$, the smallest set of registers, RAM cells, opcodes, or program variables whose intervention reproduces y, to the size $|U_{\hat{E}}|$ of the cause set the explanation actually names. An explanation that names exactly the minimal set scores 1; one that records the whole machine state pays for every spurious cell. The target level is the algorithmic unit, the register, RAM cell, opcode, module, or program variable, not the pixel. Whole-state recording names all 128 RAM cells, where the true cause set averages a handful. It therefore scores $\mathrm{M}_{\mathrm{score}} = 0.07$ across the scored games. Its faithfulness is perfect because it withholds nothing, but it is maximally non-minimal.

Two reporting axes carry the headline comparison, faithfulness against the ground truth and a plausibility proxy. The plausibility proxy is a documented, tradition-level value, not a human-subjects measurement, and we report it as such. It is assigned per method tradition, not per method instance, from the published evidence on how convincing each tradition's explanations are taken to be, before any faithfulness score on our machine is computed, so the proxy cannot be tuned to the result. The per-tradition assignment, its rubric, and its sources are tabulated in the Supplement (Section S7). There a sensitivity analysis perturbs the proxy under additive, rank-preserving, rank-jittering, and leave-one-tradition-out schemes. It finds the faithfulness-versus-proxy rank correlation stays negative throughout (from -0.82 to -0.10), so the headline contrast survives every alternative proxy assignment we tried. The proxy lets us find explanations with high plausibility but low faithfulness: explanations that convince but are wrong. F is satisfied by the wiring; S and M must be satisfied by the *computation*. Separating them lets us measure, later, what faithfulness alone leaves out.

3.5 Methods under test, games, and aggregation

We apply three traditions to the machine and score each by the triad of Eq. (3). The neuroscience analyses (Phase A) run connectomics and data-flow recovery, single-unit lesions, tuning curves, pairwise and global correlation structure, pooled-activity spectra, Granger causality, dimensionality reduction, and whole-state recording, scoring each against the known mechanism to quantify Kording's lesson at the low-faithfulness baseline [4]. The attribution and XAI methods (Phase B) run gradient saliency and its variants, integrated gradients, occlusion, perturbation masks, randomized masking, LIME, and Shapley-style attribution [5, 6, 11–15]; Grad-CAM and its variants and attention rollout [9, 10, 41] require neural layers or a learned policy and are recorded as not applicable. The mechanistic-interpretability methods (Phase C) run activation patching and causal mediation, interchange interventions and distributed alignment search, attribution and path patching, automatic circuit discovery, sparse autoencoders, causal scrubbing, linear probing with control tasks, and the logit and tuned lens [17–27, 43].

Aggregate faithfulness scoring runs on the 44 scored games, the T3-labeled roster minus the 10 games documented in Supplement S3 that fail the cause-density gate or carry no verified moving sprite, where the oracle's re-runs are exact. The six core games of Section 3.3 remain the running examples and carry the full tiered ground truth and oracle. Cell- and pixel-level scores, correlation, deletion and insertion area-under-curve, and precision at k, are taken against T1 and need no T3; only object- and concept-level scores draw on the external T3 labels, and we flag every such score as alignment-to-given, not discovery. We aggregate the committed per-game records into one row per method on the shared faithfulness-versus-plausibility axes of the Results leaderboard. The headline contrast is the all-regime gap. Pooled over every output, causal and intervention methods average a faithfulness of 0.71 against 0.38 for gradient and correlational methods, a gap of 0.32 whose 95% bootstrap interval $[0.30, 0.35]$ excludes zero. On the position regime alone, where the naive gradient is zero under the hard forward map by Prop. 1, the two families average 0.56 against 0.22, a gap of 0.34. These position-regime scores already hand the gradient-based

methods the bilinear sampler of the shared testbed; under the naive gradient those methods score 0 (Section 1.4). This gap is now statistically significant: a bootstrap over the 44 games gives a 95% interval of $[0.29, 0.38]$ that excludes zero. The per-method extreme places an exact causal method, activation patching, at faithfulness 1.000 on the position regime and the field's default saliency at only 0.092 there, the popular method nonetheless carrying the higher plausibility proxy (0.9 versus 0.5). We measure the semantic gap relative to these numbers.

Data availability. The interpretability scores reported here come from the committed analysis records in the repository. These records hold the per-method, per-game R results and the aggregated cross-tradition leaderboard [35, 66]. Every quantitative claim in the Article reads from one of these committed artifacts. No number is rerun or estimated. The game cartridges used as the subject are third-party Atari 2600 ROMs that we do *not* redistribute. We follow the practice of the underlying emulator port [30]. We acquire the ROMs through the `AutoROM` retrieval procedure and report a SHA-256 hash for each ROM in the Supplementary Information. This lets others reproduce and verify the exact binaries independently. The recorded state trajectories used to train the sparse autoencoders and to run the patching and probing experiments are regenerated deterministically from those ROMs and the published action streams.

Code availability. The interpretability benchmark introduced here is available for review at https://github.com/akmaier/UnderstandingVCS. It comprises the scoring suite, the $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ triad metrics, the intervention oracle, and the per-phase method implementations. The repository also holds the reproducibility appendix that maps every reported number to its committed record and run command. A versioned snapshot will be archived on Zenodo with a minted DOI. An optional Code Ocean capsule reproduces the leaderboard and the figures from the committed result records alone. The benchmark rests on a differentiable, bit-exact emulator. It has two interchangeable versions (`jutari` in Julia, `jaxtari` in JAX/Python), and both are already public. The emulator is described in the companion port paper [30] (arXiv:2606.22447), and its source is openly available at the same repository. The companion paper supplies the bit-exactness validation and the gradient caveat that the present work imports. What is new here is the interpretability benchmark, the oracle scoring, and the semantic-gap diagnosis. We release the code under a permissive open-source licence so that a third party can score a new method against the same ground-truth oracle.

Author contributions. A.M. conceived the study, designed the ground-truth audit and the $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ triad, implemented the benchmark and oracle scoring, and wrote the manuscript. S.B. contributed to the method design and the analysis and revised the manuscript. P.K. contributed to the differentiable substrate and oracle and to the gradient-faithfulness analysis, and helped develop the differentiability framing. All authors discussed the results, interpreted the semantic-gap finding, and approved the final manuscript.

Competing interests. The authors declare no competing interests.

Ethics and broader impact. This study uses no human-subjects data; the plausibility axis is a documented plausibility proxy by method tradition rather than a measurement, and we report it as such. The subject is a fully deterministic, hand-coded program with no learned policy, so no agent is trained. Interpretability methods are dual-use: a tool that audits a system can also be used to probe it. Our contribution is a calibration that makes explanation claims more trustworthy, and we release it openly to that end. The game cartridges are not redistributed.

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References

[1] Greydanus, S., Koul, A., Dodge, J. & Fern, A. *Visualizing and understanding Atari agents. Proceedings of the 35th International Conference on Machine Learning (ICML)*, Vol. 80, 1792–1801 (PMLR, 2018).

[2] Jacovi, A. & Goldberg, Y. *Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 4198–4205 (Association for Computational Linguistics, 2020).

[3] Adebayo, J. *et al. Sanity checks for saliency maps. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 31, 9505–9515 (2018).

[4] Jonas, E. & Kording, K. P. Could a neuroscientist understand a microprocessor? *PLOS Computational Biology* **13**, e1005268 (2017).

[5] Simonyan, K., Vedaldi, A. & Zisserman, A. *Deep inside convolutional networks: Visualising image classification models and saliency maps. International Conference on Learning Representations (ICLR) Workshop* (2014).

[6] Sundararajan, M., Taly, A. & Yan, Q. *Axiomatic attribution for deep networks. Proceedings of the 34th International Conference on Machine Learning (ICML)*, Vol. 70, 3319–3328 (PMLR, 2017).

[7] Smilkov, D., Thorat, N., Kim, B., Viégas, F. & Wattenberg, M. *Smoothgrad: Removing noise by adding noise. ICML Workshop on Visualization for Deep Learning* (2017).

[8] Springenberg, J. T., Dosovitskiy, A., Brox, T. & Riedmiller, M. *Striving for simplicity: The all convolutional net. International Conference on Learning*

Representations (ICLR) Workshop (2015).

[9] Selvaraju, R. R. *et al.* Grad-CAM: Visual explanations from deep networks via gradient-based localization. *International Journal of Computer Vision* **128**, 336–359 (2020). Originally arXiv:1610.02391 (2016).

[10] Chattopadhyay, A., Sarkar, A., Howlader, P. & Balasubramanian, V. N. *Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks. IEEE Winter Conference on Applications of Computer Vision (WACV)*, 839–847 (IEEE, 2018).

[11] Zeiler, M. D. & Fergus, R. *Visualizing and understanding convolutional networks. European Conference on Computer Vision (ECCV)*, 818–833 (2014).

[12] Fong, R. C. & Vedaldi, A. *Interpretable explanations of black boxes by meaningful perturbation. IEEE International Conference on Computer Vision (ICCV)*, 3429–3437 (2017).

[13] Petsiuk, V., Das, A. & Saenko, K. *RISE: Randomized input sampling for explanation of black-box models. British Machine Vision Conference (BMVC)* (2018).

[14] Ribeiro, M. T., Singh, S. & Guestrin, C. *"why should i trust you?": Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144 (2016).

[15] Lundberg, S. M. & Lee, S.-I. *A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems (NeurIPS)* (2017).

[16] Such, F. P. *et al. An Atari model zoo for analyzing, visualizing, and comparing deep reinforcement learning agents. Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI)*, 3260–3267 (2019).

[17] Vig, J. *et al. Investigating gender bias in language models using causal mediation analysis. Advances in Neural Information Processing Systems (NeurIPS)* (2020).

[18] Meng, K., Bau, D., Andonian, A. & Belinkov, Y. *Locating and editing factual associations in GPT. Advances in Neural Information Processing Systems (NeurIPS)* (2022).

[19] Wang, K., Variengien, A., Conmy, A., Shlegeris, B. & Steinhardt, J. *Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. International Conference on Learning Representations (ICLR)* (2023).

[20] Conmy, A., Mavor-Parker, A. N., Lynch, A., Heimersheim, S. & Garriga-Alonso, A. *Towards automated circuit discovery for mechanistic interpretability. Advances in Neural Information Processing Systems (NeurIPS)* (2023).

[21] Geiger, A., Lu, H., Icard, T. & Potts, C. *Causal abstractions of neural networks. Advances in Neural Information Processing Systems (NeurIPS)* (2021).

[22] Geiger, A., Wu, Z., Potts, C., Icard, T. & Goodman, N. D. *Finding alignments between interpretable causal variables and distributed neural representations. Proceedings of the Third Conference on Causal Learning and Reasoning (CLeaR)* (2024).

[23] Cunningham, H., Ewart, A., Riggs, L., Huben, R. & Sharkey, L. Sparse autoencoders find highly interpretable features in language models (2023). 2309.08600.

[24] Alain, G. & Bengio, Y. *Understanding intermediate layers using linear classifier probes. International Conference on Learning Representations (ICLR) Workshop* (2017).

[25] Hewitt, J. & Liang, P. *Designing and interpreting probes with control tasks. Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2733–2743 (2019).

[26] Belrose, N. *et al.* Eliciting latent predictions from transformers with the tuned lens (2023). 2303.08112.

[27] Nanda, N. Attribution patching: Activation patching at industrial scale. https://www.neelnanda.io/mechanistic-interpretability/attribution-patching (2023).

[28] Atrey, A., Clary, K. & Jensen, D. *Exploratory not explanatory: Counterfactual analysis of saliency maps for deep reinforcement learning. International Conference on Learning Representations (ICLR)* (2020).

[29] Karvonen, A. *et al. SAEBench: A comprehensive benchmark for sparse autoencoders in language model interpretability. Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Vol. 267 of *PMLR* (2025). Eight metrics over >200 open-sourced SAEs; finds gains on proxy metrics do not reliably translate to practical interpretability., 2503.09532.

[30] Maier, A., Bayer, S. & Krauss, P. A differentiable Atari VCS: A complex, fully known ground truth for explainable AI. arXiv:2606.22447; code at https://github.com/akmaier/UnderstandingVCS (2026). Companion port paper (Paper 1) of this program; submitted 2026-06-21. Code: https://github.com/akmaier/UnderstandingVCS. Citation key `maier2025vcs` retained for stability., 2606.22447.

[31] DeepMind. Xitari: An Arcade Learning Environment fork. https://github.com/google-deepmind/xitari (2016). Accessed: 2026-06-16.

[32] Anand, A. *et al. Unsupervised state representation learning in atari. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32 (2019).

[33] Delfosse, Q., Blüml, J., Gregori, B., Sztwiertnia, S. & Kersting, K. OCAtari: Object-centric Atari 2600 reinforcement learning environments (2023). 2306.08649.

[34] IEEE. IEEE standard glossary of software engineering terminology (ieee std 610.12-1990). IEEE, New York (1990). Defines "software" as computer programs, procedures, and possibly associated documentation and data.

[35] Paper 2 authors. Cross-tradition faithfulness–plausibility leaderboard for the differentiable Atari VCS interpretability audit (phases a–c). Committed analysis record, `tools/xai_study/compare/out/leaderboard.json`; aggregates the per-game R records under `tools/xai_study/phaseA_kording/out/` (2026). Internal data artifact; 1773 per-game records aggregated into 31 method rows including the oracle positive control. Generated by `compare/leaderboard.py`. Accessed 2026-06-23.

[36] Shrikumar, A., Greenside, P. & Kundaje, A. *Learning important features through propagating activation differences. Proceedings of the 34th International Conference on Machine Learning (ICML)*, 3145–3153 (2017).

[37] Erion, G., Janizek, J. D., Sturmfels, P., Lundberg, S. M. & Lee, S.-I. Improving performance of deep learning models with axiomatic attribution priors and expected gradients. *Nature Machine Intelligence* **3**, 620–631 (2021).

[38] Štrumbelj, E. & Kononenko, I. Explaining prediction models and individual predictions with feature contributions. *Knowledge and Information Systems* **41**, 647–665 (2014).

[39] Fong, R., Patrick, M. & Vedaldi, A. *Understanding deep networks via extremal perturbations and smooth masks. Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2950–2958 (2019).

[40] Olson, M. L., Khanna, R., Neal, L., Li, F. & Wong, W.-K. Counterfactual state explanations for reinforcement learning agents via generative deep learning. *Artificial Intelligence* **295**, 103455 (2021).

[41] Abnar, S. & Zuidema, W. *Quantifying attention flow in transformers. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 4190–4197 (2020).

[42] Bastani, O., Pu, Y. & Solar-Lezama, A. *Verifiable reinforcement learning via policy extraction. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 31, 2494–2504 (2018).

[43] Chan, L. *et al.* Causal scrubbing: A method for rigorously testing interpretability hypotheses. AI Alignment Forum (2022).

[44] Mueller, A. *et al. MIB: A mechanistic interpretability benchmark. Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Vol. 267 of *PMLR* (2025). Two tracks (circuit localization; causal-variable localization) over four tasks and five neural language models., 2504.13151.

[45] Li, X. *et al.* $\mathcal{M}^4$*: A unified XAI benchmark for faithfulness evaluation of feature attribution methods across metrics, modalities and models. Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, Vol. 36 (2023).

[46] Balduzzi, D. *et al. The shattered gradients problem: If resnets are the answer, then what is the question? Proceedings of the 34th International Conference on Machine Learning (ICML)*, Vol. 70 of *PMLR*, 342–350 (2017).

[47] Morcos, A. S., Barrett, D. G. T., Botvinick, M. & Rabinowitz, N. C. *On the importance of single directions for generalization. International Conference on Learning Representations (ICLR)* (2018). 1803.06959.

[48] Leavitt, M. L. & Morcos, A. S. *Towards falsifiable interpretability research. NeurIPS 2020 Workshop: ML Retrospectives, Surveys & Meta-Analyses (ML-RSA)* (2020). 2010.12016.

[49] Elhage, N. *et al.* Toy models of superposition (2022). Transformer Circuits Thread, https://transformer-circuits.pub/2022/toy_model/index.html, 2209.10652.

[50] Kindermans, P.-J. *et al.* in *The (un)reliability of saliency methods Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*, Vol. 11700 of *Lecture Notes in Computer Science* 267–280 (Springer, 2019).

[51] Sturmfels, P., Lundberg, S. & Lee, S.-I. Visualizing the impact of feature attribution baselines. *Distill* **5** (2020). https://distill.pub/2020/attribution-baselines.

[52] Jain, S. & Wallace, B. C. *Attention is not explanation. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 3543–3556 (Association for Computational Linguistics, 2019).

[53] Lindner, D. *et al. Tracr: Compiled transformers as a laboratory for interpretability. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36 (2023). 2301.05062.

[54] Gupta, R., Arcuschin, I., Kwa, T. & Garriga-Alonso, A. *InterpBench: Semi-synthetic transformers for evaluating mechanistic interpretability techniques.*

Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, Vol. 37 (2024). 2407.14494.

[55] Yang, M. & Kim, B. BAM: Benchmarking attribution methods with relative feature importance (2019). Introduces the BAM benchmark: models trained with known relative feature importance plus paired-model/paired-input metrics for attribution methods. (Key `yang2019bim` retained for citation stability; the work is BAM, not BIM.), 1907.09701.

[56] Bellemare, M. G., Naddaf, Y., Veness, J. & Bowling, M. The Arcade Learning Environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research* **47**, 253–279 (2013).

[57] Machado, M. C. *et al.* Revisiting the Arcade Learning Environment: Evaluation protocols and open problems for general agents. *Journal of Artificial Intelligence Research* **61**, 523–562 (2018).

[58] Dalton, S., Frosio, I. & Garland, M. *Accelerating reinforcement learning through GPU Atari emulation. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33 (2020). ArXiv:1907.08467; code at https://github.com/NVlabs/cule.

[59] Delfosse, Q. *et al.* JAXAtari: A high-performance framework for reasoning agents. https://github.com/k4ntz/JAXAtari (2026). Software; accessed 2026-06-19.

[60] Marr, D. *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information* (W. H. Freeman and Company, 1982).

[61] Lazebnik, Y. Can a biologist fix a radio? — or, what i learned while studying apoptosis. *Cancer Cell* **2**, 179–182 (2002).

[62] Barbiero, P. *et al.* Neural interpretable reasoning. *arXiv preprint arXiv:2502.11639* (2025).

[63] Bengio, Y., Léonard, N. & Courville, A. Estimating or propagating gradients through stochastic neurons for conditional computation (2013). 1308.3432.

[64] Jang, E., Gu, S. & Poole, B. *Categorical reparameterization with Gumbel-softmax. International Conference on Learning Representations (ICLR)* (2017).

[65] Jaderberg, M., Simonyan, K., Zisserman, A. & Kavukcuoglu, K. *Spatial transformer networks. Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 28 (2015).

[66] Paper 2 authors. Faithful-method demonstration: causal near-ceiling versus popular near-chance on the differentiable Atari VCS. Committed analysis record,

`tools/xai_study/compare/out/faithful_demo.md`; pure read over the committed cross-tradition leaderboard (`compare/out/leaderboard.json`) and the per-game R records it aggregates. Generated by `compare/faithful_demo.py`. (2026). Internal data artifact. Position-regime pair activation_patching = 1.000 vs vanilla_saliency = 0.000 on all six core games (per-method gap 1.000). Accessed 2026-06-23.

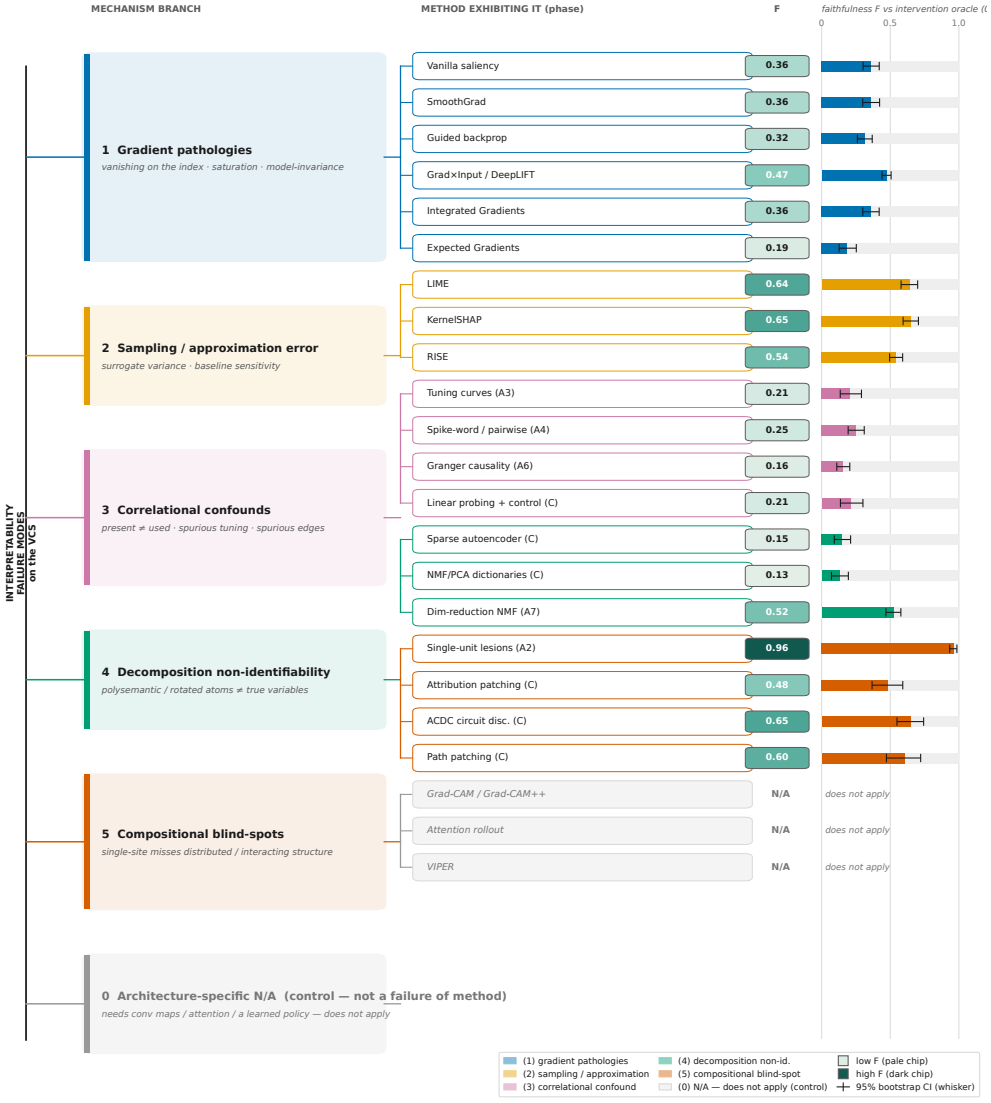


Fig. 4 A taxonomy of interpretability failure on the VCS, organised by mechanism. Each leaf is a method exhibiting the branch's failure mode, annotated with its *family-mean* faithfulness $F \pm 95\%$ bootstrap-over-games CI against the intervention oracle (Section 3.2; 0 = chance, 1 = oracle ceiling); the per-game exemplars quoted below carry their own "on ⟨game⟩" label. The lower branches confirm the expected failures: the naive gradient is provably zero on the discrete position/index regime (the family's position-regime mean is only 0.22, even with the sampler-restored gradient included), and the architecture-specific methods (Grad-CAM, attention, VIPER) do not apply because the VCS lacks the neural substrate they require, a property of the subject, not a method failure. The higher-order branches are the *semantic gap*: present-but-unused concepts (linear probing [25] flags a present-but-unused cell the oracle says is unused), named-but-unused factorisations (on Pong the SAE [23] matches every varying variable at fraction 1.0 yet scores $F = 0.041$), and graph-correct-but-incomplete circuits (ACDC [20] at $F = M = 1.0$, $S = 0.10$ on Q*bert). The neural-network parallels shown alongside each branch are documented analogues [46–52], evidence transfer rather than proof. F is a point estimate and the whisker a 95% bootstrap CI over games; only the failure-exhibiting subset is drawn here, out of the full 31-row leaderboard (30 methods plus the oracle). Every annotated value reads from a committed record.

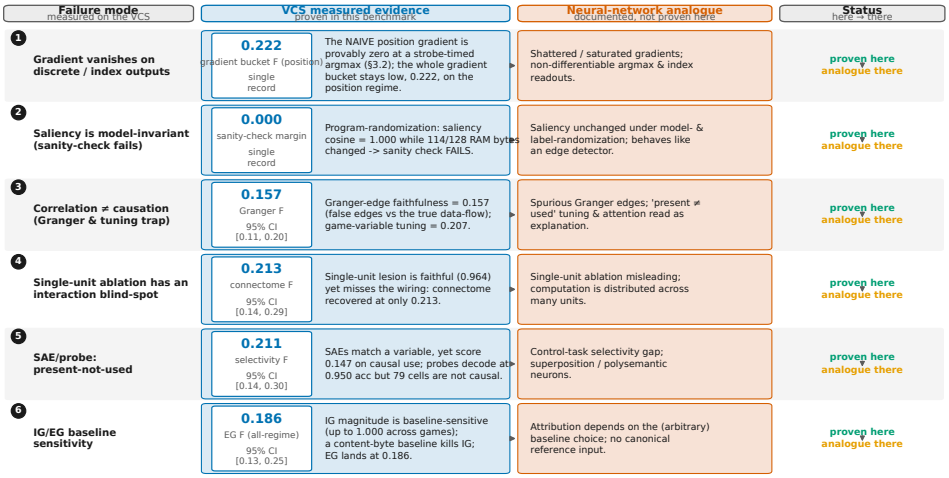


Fig. 5 The VCS↔neural-network failure-mode map. Each row pairs a failure mode we measured on the VCS against the oracle (Section 3.2) — left, blue — with the neural-network behavior it is a documented analogue of — right, vermillion. On the VCS the failure is provable because we hold the exact ground truth; the neural-network column is a documented analogue, not proven by this benchmark, and reads the attribution and probing literature ([3, 6, 46, 51] on gradients; [25, 47, 48, 52] on use vs. presence; [32, 49, 50] on probing, superposition, and reliability). The takeaway is one line: every VCS failure has a named counterpart in how interpretability methods are reported to fail on networks, so a property provable here is one to suspect there. On the position regime the causal and intervention methods open a 0.338 faithfulness gap over the gradient and correlational methods, whose 95% bootstrap interval over the 44 games $[0.29, 0.38]$ excludes zero, so the advantage is now statistically significant; the all-regime gap of 0.32 likewise excludes zero. The naive gradient is provably zero on the position read, and even after the sampler restores a gradient the position-regime mean stays low (about 0.22 for the gradient family); activation patching, by contrast, reaches the oracle ceiling (1.000) there where vanilla saliency reaches only 0.092. Per-row VCS scores, their committed records, and the full map are in the Supplementary Information.

Supplementary Information

This Supplementary Information records what the main paper summarizes. We give the per-method protocols (Section S1) and the leaderboard schema with its glossary (Section S2). We give the coverage and applicability of every method across games and output regimes (Section S3). We map each headline claim to its evidence (Section S4). We trace the provenance of every headline number down to the record file and command that produced it (Section S5). We state plainly what the paper does not claim (Section S6). We give the plausibility-proxy rubric and its sensitivity analysis (Section S7). We list a SHA-256 hash for each ROM in the bit-exact roster (Section S8), so the exact game binaries can be verified without our redistributing them. The correctness triad $\mathrm{F} \wedge \mathrm{S} \wedge \mathrm{M}$ used throughout is the one defined in the main Methods (Section 3.4); we reference it here rather than restating it. The benchmark scores 30 interpretability methods together with the intervention oracle (Section 3.2) as a positive control, for 31 leaderboard rows in total.

S1 Per-method protocols

This section records five things for each method under test. It records the exact objects the method consumes and produces, the reference it is scored against, its sampling or optimization budget, the metric that turns its output into a faithfulness score, and the committed record file (version-controlled, in the released repository) that holds the per-game results. All methods share one evaluation contract. We state that contract once and then specialize it per method. Every method receives the same bit-exact VCS state on each game of the scored set. We state the contract on the six core games (`breakout`, `ms_pacman`, `pong`, `qbert`, `seaquest`, `space_invaders`), which carry the full tiered ground truth and the exact oracle and serve as the running example; the aggregate leaderboard scores extend the identical protocol to the 44-game scored set of Section S3.2. That state is not the boot screen. It is a shared gameplay state, produced by a single seeded random-action stream (seed 0) run for a 90-frame prefix and then rolled forward for a 15-frame analysis horizon. The stream is deterministic, so every method sees the identical state on each game, and the run is asserted bit-exact per game. An oracle cause-density gate screens the state before any method runs. The gate counts how many candidate causes move the output by more than 0.5 and flags a state with fewer than four. It does not re-sample the state; a game whose shared state fails the gate is left out of the scored set (Section S3.2), so a near-flat oracle column is never scored. On `seaquest` the redesigned shared state and output raise the count of live candidate causes from 2 to 19. The candidate-cause set is the machine's own state. This set is the RAM bytes together with the CPU registers and the TIA and RIOT registers exposed by the substrate. The output every method attributes over is a shared screen-buffer count read at the end of the analysis horizon, not a single hand-picked pixel. It counts the pixels of the whole frame that differ from the un-intervened frame; the record label `screen_region(n_changed_px)@r·c·` names only the cell at which the sampler gradient is taken. The reference is the intervention oracle of Section 3.2, evaluated by exact re-runs of the emulator under $\mathrm{do}(u)$. A method that cannot be applied to a given output regime is recorded as not-applicable rather than

Table S1 Consolidated leaderboard: all methods, all triad axes. One row per method across the three phases, plus the intervention oracle as a positive control. F is faithfulness (the per-method metric against the oracle, with the bootstrap-over-games 95% CI in brackets), S is sufficiency, M is minimality, all as defined in Section 3.4. Every per-record F, S, and M is clipped to $[0, 1]$ before averaging, while the per-game records keep the raw value; Plaus is the tradition-level plausibility proxy and n is the number of scored games. A5 (field potentials) and A7 (dimensionality reduction) lack the S/M decomposition, and path patching, the logit lens, and linear probing carry no S; these cells are marked "—". Values are transcribed from `tools/xai_study/compare/out/leaderboard.json` (the committed numbers of record).

Method	Phase	Tradition	F (95% CI)	S	M	Plaus	n
A1 connectomics	A	intervention	0.213 [0.138, 0.289]	0.957	0.971	0.55	38
A2 lesions	A	intervention	0.964 [0.933, 0.986]	0.610	1.000	0.55	44
A3 tuning curves	A	correlational	0.207 [0.135, 0.290]	0.196	0.918	0.80	44
A4 spike-word / pairwise	A	correlational	0.251 [0.193, 0.310]	0.147	0.393	0.80	36
A5 field potentials	A	correlational	0.394 [0.341, 0.449]	—	—	0.80	42
A6 Granger	A	correlational	0.157 [0.110, 0.205]	0.878	0.701	0.80	38
A7 dim. reduction	A	dim. red.	0.523 [0.468, 0.577]	—	—	0.75	44
A8 whole-state	A	descriptive	1.000 [1.000, 1.000]	1.000	0.075	0.30	44
Vanilla saliency	B	gradient	0.361 [0.302, 0.420]	0.555	0.980	0.90	44
smoothgrad	B	gradient	0.361 [0.300, 0.422]	0.558	0.982	0.90	44
Guided backprop	B	gradient	0.315 [0.260, 0.368]	0.546	0.966	0.90	44
Grad×Input / DeepLIFT	B	gradient	0.471 [0.440, 0.506]	0.604	0.993	0.90	44
Integrated Gradients	B	gradient	0.361 [0.300, 0.419]	0.553	0.985	0.90	44
Expected Gradients	B	gradient	0.186 [0.127, 0.252]	0.548	1.000	0.90	44
kernelshap	B	gradient	0.650 [0.594, 0.705]	0.637	0.565	0.90	44
lime	B	gradient	0.640 [0.579, 0.700]	0.646	0.553	0.90	44
rise	B	gradient	0.544 [0.496, 0.590]	0.563	0.356	0.90	44
occlusion	B	intervention	0.685 [0.627, 0.740]	0.654	0.972	0.55	44
Extremal perturbation	B	intervention	0.458 [0.406, 0.514]	0.608	0.968	0.55	44
On-distribution counterfactual	B	intervention	0.433 [0.346, 0.514]	0.610	0.986	0.55	44
Activation patching	C	causal	1.000 [1.000, 1.000]	0.996	0.964	0.50	44
Causal scrubbing	C	causal	0.997 [0.992, 1.000]	0.997	0.800	0.50	44
Interchange interventions (DAS)	C	causal	1.000 [1.000, 1.000]	1.000	0.869	0.50	44
Logit / tuned lens	C	causal	0.793 [0.731, 0.847]	—	0.191	0.50	44
Path patching	C	causal	0.603 [0.472, 0.722]	—	1.000	0.50	44
ACDC	C	causal	0.648 [0.549, 0.743]	0.314	1.000	0.50	38
Attribution patching	C	gradient	0.482 [0.367, 0.591]	0.798	0.990	0.90	44
Linear probing (+ control)	C	probing	0.211 [0.137, 0.301]	—	0.944	0.85	26
NMF / PCA dictionaries	C	dim. red.	0.129 [0.072, 0.194]	0.725	0.985	0.75	44
Sparse autoencoder	C	dim. red.	0.147 [0.092, 0.211]	0.165	0.924	0.75	42
Intervention oracle	—	oracle	1.000 [1.000, 1.000]	1.000	1.000	1.00	1

scored as zero. An empty result is therefore never read as a wrong one. Faithfulness is the F axis of the main-text triad (Section 3.4); we reference that definition and do not restate it. Records live under `tools/xai_study/<phase>/out/`; the aggregate they feed is `tools/xai_study/compare/out/leaderboard.json`. Table S1 consolidates the triad scores for all thirty methods and the oracle positive control in one place; the per-method protocols follow.

The neuroscience analyses (Phase A) treat the running machine as a brain. It asks how far each classical analysis recovers the known mechanism on the shared gameplay state. Each method's baseline is the exact data-flow graph or causal role that the substrate makes available. Its budget is the scored set (Section S3.2) unless noted, with the six-game core as the running example, and its scoring metric is named in Table S2. One core cell needs a note. An earlier pairwise-analysis (A4) record on `ms_pacman`, produced on its own trajectory before the shared testbed existed, has been removed.

Table S2 Phase A protocols (neuroscience analyses). One row per method. The columns give the input it reads, the reference it is scored against, the sampling or intervention budget, the scoring metric, and the committed record. All eight run on the scored set (Section S3.2), with the six core games as the running example; faithfulness is the F axis of Section 3.4. The whole-state recording (A8) is included as a descriptive upper bound on coverage. Its faithfulness F is trivial, so we read it for its minimality M.

Method	Input / candidate causes	Reference	Scoring metric	Record
A1 connectomics	RAM/register activity over trajectory	data-flow graph	data-flow graph F_1 vs. oracle	A1
A2 lesions	single-cell knockouts via do(u)	true causal role	lesion-importance ρ vs. role	A2
A3 tuning curves	per-cell response vs. variable	oracle causal importance	rank corr. of tuning selectivity vs. importance	A3
A4 spike-word / pairwise	pairwise cell correlations	true coupling	corr. ρ vs. coupling	A4
A5 field potentials	pooled-activity spectra	true causal share	$1-$ clock-explained var.	A5
A6 Granger	directed lag structure (lag 2, $\alpha = 0.05$)	true data-flow	edge F_1 vs. data-flow	A6
A7 dim. reduction	NMF/PCA over activity	known variables	NMF matched frac. of top-5 components (purity ≥ 0.5)	A7
A8 whole-state	full state record	oracle decomp.	minimality M vs. oracle	A8

Records: `tools/xai_study/phaseA_kording/out/A{1..8}_*`; all eight run on the 44-game scored set (Section S3.2), with the six core games as the running example (`n_games` per method in `leaderboard.json`).

At the shared testbed the A4 runner finds no coupled candidate pair on `ms_pacman` and does not score it, as for the other games A4 does not score; A4 is scored on 36 of the 44 games. The other five core games carry A4 on the shared state.

The attribution methods (Phase B) run the everyday XAI toolbox over the same gameplay state. These methods produce a real-valued saliency or attribution over the candidate-cause set. We score that output by Pearson correlation against the oracle attribution. The gradient family runs with the bilinear sub-pixel sampler turned on, so it is scored on a real position gradient and not on the naive one. The naive gradient of the discrete position output is provably zero (Prop. 1); the sampler restores a finite gradient, but the restored gradient is low and mixed in faithfulness rather than a fix. Expected Gradients draws its reference pool from the states of the same seeded gameplay trajectory, not from a synthetic baseline. Two patterns are degenerate rather than informative. The on-distribution counterfactual returns a zero delta on both outputs of `space_invaders`, `ms_pacman`, `qbert`, `gopher`, `kangaroo`, `kung_fu_master` and `pacman`, and on the content output of `enduro`, where its perturbation does not

move the shared output at that frame; those cells record $F = 0$ and enter the mean at that value. Expected Gradients scores low on the content regime of every core game except `pong`, because the causal byte it should land on is static there. The end-to-end gradient control of Section 1.2 (`e2e_gradient.jl`, all 44 scored games, records `e2e_{saliency,gradxinput}_⟨game⟩_*`, merged in `e2e_core_summary.json` and compared with the single read in `compare/out/e2e_vs_single.json`) is not a separate leaderboard row. It differentiates the end-of-horizon output back to the RAM at the intervention frame through the full-system soft step, one instruction at a time (57,073 to 106,547 instructions per game). Reverse mode is checkpointed every 200 instructions, and each chunk's soft forward pass is asserted bit-equal to the hard registers and RAM. The position output chains the rollout with the same bilinear sampler at the end of the horizon. It is scored with the same oracle, causes, and metrics as vanilla saliency. The backward pass takes 8–23 h per game on one CPU core. On the six core games two independent runs (ARM and x86) agree to within one unit in the last place. The budgets that matter are the Monte-Carlo or perturbation counts of the sampling estimators, recorded per method in Table S3. The seed-dispersion of the stochastic members is carried in `leaderboard_ci.csv` (`seed_sd`, `seed_sd_source`) and reproduced in Section S5.

The mechanistic methods (Phase C) are built to recover circuits. Each is scored against the true data-flow or the exact recovered-minus-exact deviation. The causal members of this phase are the ones that reach the oracle ceiling. We list their exact scoring metric so that the ceiling is not read as a tautology. Each member is scored against the substrate's known routine, and not against another method. Two members (NMF/PCA dictionaries and the sparse autoencoder) are dimensionality-reduction methods carried here for comparison. The sparse autoencoder has two aggregations that differ, which we reconcile in Section S5.

The intervention oracle is the positive control, not a method under test. It is the exact do(u) ground truth of Section 3.2, evaluated by re-running the bit-exact machine. It attains $F = S = M = 1.0$ by construction. Its record triple is the intervention oracle, the content-path gradient oracle, and their cross-check (`tools/xai_study/ground_truth/out/`). It anchors the ceiling against which every other row is read, which is why it occupies the thirty-first leaderboard row.

S2 Benchmark schema and glossary

The leaderboard is built from a flat record schema. Stating it removes the ambiguity about what a single number on the benchmark counts. Each method is run per game and writes a per-game record under its phase's `out/` tree. The leaderboard aggregates those records into one row per method by averaging over games. The schema columns of a leaderboard row are given in Table S5. The aggregation rule is a mean over the n_{games} games. When a method emits several output records on one game, those records are first reduced within that game. Every per-record F, S, and M is clipped to $[0, 1]$ before this averaging; the per-game records keep the raw value. The faithfulness, sufficiency, and minimality columns report the F, S, and M axes defined in Section 3.4. We do not re-derive those definitions here.

Table S3 Phase B protocols (attribution methods). Each method emits a saliency over the candidate-cause set. We score it by Pearson correlation against the oracle attribution (Section 3.2). The reference is shared and the budget is the scored set of Section S3.2 (six core games as the running example). The budget column records the sampling or perturbation count that drives each estimator. The seed column flags which methods carry an independent-seed dispersion in `leaderboard_ci.csv`.

Method	Tradition	Budget / hyperparameters	Seed sd
Vanilla saliency	gradient	single backward pass	—
smoothgrad	gradient	noise-averaged gradient; σ-robustness committed	0.0
Guided backprop	gradient	single guided pass	—
Grad×Input / DeepLIFT	gradient	DeepLIFT reference-difference	—
Integrated Gradients	gradient	path integral, baseline = zero state	—
Expected Gradients	gradient	expectation over baselines (refits)	0.0
kernelshap	gradient	Monte-Carlo coalition sampling ($\propto 1/\sqrt{N}$)	0.019
lime	gradient	local linear fit, perturbation refits	0.005
rise	gradient	random-mask Monte-Carlo ($\propto 1/\sqrt{N}$)	0.061
occlusion	intervention	exhaustive single-cell masking	—
Extremal perturbation	intervention	optimized mask, area-constrained	—
On-distribution counterfactual	intervention	on-manifold do(u) resampling	—

Records: `tools/xai_study/phaseB_attribution/out/`; all twelve run on the 44-game scored set (Section S3.2), six core games as the running example, and are scored by Pearson correlation against the oracle (Section 3.2). Seed sd is the independent-seed dispersion from `leaderboard_ci.csv`.

We fix the vocabulary that the schema implies, because the review asked for a single unambiguous method count. A *method* is one interpretability technique, e.g. `vanilla_saliency`. A *method row* is its single aggregated line on the leaderboard. A *per-game record* is one method scored on one game, written under a phase `out/` tree. An *output record* is a per-game record for one output regime; a method may emit several of these on one game. A *family mean* is the average over the members of a tradition family, e.g. causal/intervention. The *oracle positive control* is the exact do(u) ground truth of Section 3.2, scored as a method to anchor the ceiling. A *not-applicable method* is one that does not apply to a given output regime; it is recorded as N/A rather than as a zero. With this vocabulary the count is unambiguous. The benchmark scores 30 interpretability methods together with the intervention oracle as a positive control, for 31 leaderboard rows in total (`leaderboard.json.n_methods` = 31), aggregating 1862 per-game records over the 44-game scored set (`n_per_game_records_aggregated` = 1862): 8 methods in Phase A, 12 in Phase B, 10 in Phase C, and the one oracle row. This count is the canonical one used in the abstract, the figures, and throughout this Supplement.

Table S4 Phase C protocols (mechanistic methods). Each method is scored against the substrate's known circuit or the exact recovered-minus-exact deviation (Section 3.2), and not against another method. This is what keeps the oracle ceiling non-circular. The probing and dictionary-learning members are included for comparison with the causal members.

Method	Tradition	Scoring metric
Activation patching	causal	max |recovered − exact| vs. oracle
Causal scrubbing	causal	scrubbing-preserved performance
Interchange interventions (DAS)	causal	aligned interchange accuracy
Logit / tuned lens	causal	faithful-weight fraction (readout fidelity 1.0 is a positive control)
Path patching	causal	path-circuit F_1 vs. true routine
ACDC	causal	discovered-circuit best F_1 vs. data-flow
Attribution patching	gradient	corr. of approximation vs. exact patch
Linear probing (+ control)	probing	selectivity vs. control task
NMF / PCA dictionaries	dim. red.	matched-component fraction vs. vars
Sparse autoencoder	dim. red.	Spearman of per-cell recovery vs. oracle importance

Records: `tools/xai_study/phaseC_mechanistic/out/`; all ten run on the 44-game scored set (Section S3.2), with the six core games as the running example.

S3 Coverage and applicability

The benchmark scores methods on a 44-game scored set drawn from a larger bit-exact substrate, with a six-game core as the running example. The coverage table makes the scope of that choice explicit. Paper 1 validates the differentiable VCS ports bit-for-bit against the `xitari` reference on all 64 Arcade Learning Environment games, within a 30-frame RAM and 60-frame screen window. The six core games listed in Table S6 carry the full tiered ground truth; the aggregate scores extend the same protocol to the 44-game scored set of Section S3.2. Every method is scored on a shared gameplay state, produced by one seeded random-action stream (seed 0, 90-frame prefix, 15-frame analysis horizon), not on the boot screen. This gives the oracle a rich, non-degenerate cause set to score against. The scored frame (105) lies beyond the window Paper 1 certifies, which is 30 RAM and 60 screen frames on fixed action streams, so we validated the analysed streams directly: for all 44 scored games the port matches `xitari` on every frame from boot to 105, in RAM and in screen (Section S3.2). The oracle in addition asserts determinism by re-running each scored state twice. We pick these six games as the core because all three tiers of ground truth (Section 3.3) are available for them and the oracle is exact. Of the remaining 58 games, 38 more enter the scored set. The other 20 are bit-exact substrate but are not scored: 10 carry no T3 label, and 10 labeled games are held out for the two methodological reasons of Section S3.2. Section S3.1 (Table S8) records how far the external T3 labels reach across the 64-game roster, defining the 54-game labeled breadth set from which the scored set is drawn. The conformance column records, for the scored games, the frame-by-frame check of the

Table S5 Leaderboard row schema. The columns of one row in `compare/out/leaderboard.json` (`rows[]`) and its uncertainty companion `leaderboard_ci.csv`. The metric column names the per-method scoring function from Section S1; F, S, and M are the triad axes of Section 3.4. Aggregation is a mean over n_{games}, with bootstrap confidence intervals over games (and, where present, over records).

Column	Meaning
`phase`	phaseA_kording, phaseB_attribution, phaseC_mechanistic, or ground_truth
`method`	method identifier (the row key)
`tradition`	intervention, causal, gradient, correlational, `dim_reduction`, probing, descriptive, oracle
`n_games`	number of games the method was scored on (up to 44 for the scored set of Section S3.2; 1 for the oracle)
`n_records`	number of per-game output records aggregated into the row
`faithfulness` (F)	faithfulness vs. the intervention oracle (Section 3.2), $[0, 1]$, higher better
`faithfulness_ci95`	95% normal-approximation CI half-width over the output records (the bootstrap-over-games CI is in `leaderboard_ci.csv`)
`faithfulness_position_regime`	F restricted to the discrete sprite-position/index regime
`faithfulness_content_regime`	F restricted to the continuous content regime
`triad_S` / `triad_M`	sufficiency / minimality axes (Section 3.4), where applicable
`plausibility_proxy`	documented method-tradition proxy, $[0, 1]$ — not a human-subjects measurement
`is_positive_control`	1 for the oracle row, 0 otherwise
`metric_names`	the per-method scoring function (Section S1)
`commits`	the commit(s) at which the per-game records were produced

analysed stream and, for the unscored games, Paper 1's certified window and its later 3,600-frame check. The tier column records which of the three ground-truth tiers (T1 exact intervention, T2 data-flow, T3 externally-supplied semantic labels) is available.

A gameplay state can still be degenerate for a particular method, and the benchmark records this explicitly. The oracle cause-density gate is the first guard. It counts how many candidate causes move the shared output by more than 0.5 and flags a state with fewer than four. The gate does not re-sample the state; a game whose shared state is flagged is held out of the scored set (Section S3.2), so a near-flat oracle column never enters the leaderboard. On `seaquest` the shared gameplay state and output raise the count of live candidate causes at the analysis frame from 2 to 19, which is what makes the game usable. Three residual degenerate patterns survive the gate. The pairwise neuroscience analysis (A4) finds no coupled candidate pair on `ms_pacman` at the shared state, so it does not score that game, as for the other games A4 does not score (36 of 44 are scored); an earlier `ms_pacman` record produced before the shared testbed has been removed. The on-distribution counterfactual returns a zero delta on both outputs of seven scored games (the core games `space_invaders`, `ms_pacman` and `qbert`, and `gopher`, `kangaroo`, `kung_fu_master` and `pacman`) and on the content output of `enduro`, where its perturbation does not move the shared output; those cells record $F = 0$. Expected Gradients scores low on the content regime of every core game but `pong`, because the causal byte it should attribute to is static there. These last two enter the leaderboard means at their recorded values; we name them so that the low scores are read as degenerate cells rather than as evidence about the method.

Table S6 Core-game coverage. The six core games carry exact tiered ground truth and the full set of methods. "0–105" means the analysed stream (the seed-0 testbed stream the methods score) matches `xitari` on every frame from boot to frame 105, in RAM and in screen (Section S3.2). "30/60; 3600" is the certified Paper 1 window (30 RAM / 60 screen frames) and the later 3,600-frame check on Paper 1's fixed action streams. [a]That check has transient exceptions in `alien` and `carnival` (screen) and `amidar` (RAM); none falls inside frames 0–105 of the analysed streams. The last two columns give the methods run and the per-game output records contributed. A further 38 games complete the 44-game scored set (Section S3.2); the remaining 20 games of the bit-exact 64-game substrate (Paper 1) are not scored. T3 labels are external (Section 3.3); we record this so the benchmark does not appear to certify them.

Game	Paper-1 status	Conformance	Tiers	Methods run	Records
breakout	bit-exact	0–105	T1,T2,T3	30 + oracle	per-game
ms_pacman	bit-exact	0–105	T1,T2,T3	30 + oracle	per-game
pong	bit-exact	0–105	T1,T2,T3	30 + oracle	per-game
qbert	bit-exact	0–105	T1,T2,T3	30 + oracle	per-game
seaquest	bit-exact	0–105	T1,T2,T3	30 + oracle	per-game
space_invaders	bit-exact	0–105	T1,T2,T3	30 + oracle	per-game
38 further scored games	bit-exact	0–105	T1,T2,T3	up to 30 + oracle	per-game
20 unscored games	bit-exact	30/60; 3600[a] fr	T1,T2 partial	not scored	—

Whether a method applies at all depends on what the method needs from its target. The applicability matrix records that for every method family. The Phase A neuroscience analyses read the raw machine state and need neither a learned policy nor gradients. The Phase B attribution methods split into two groups. The gradient members need a differentiable forward pass. The intervention members need only the ability to set state. The Phase C mechanistic methods need access to internal activations. The causal members among them also need the ability to intervene. Table S7 records, per family, whether the method operates on the raw VCS, whether it requires a learned policy, gradients, internal layers or attention, or external labels, whether it uses oracle-like interventions, and whether it receives a supplied hypothesis. The table shows the same pattern the results confirm. The families that score high on faithfulness are the ones that can intervene. The families that collapse on the position regime are the gradient and correlational ones.

S3.1 T3 semantic-label coverage across the 64-game roster

The tier-3 (T3) ground truth is externally supplied: it consists of semantic object and position labels imported from AtariARI [32] and OCAtari [33], not derived by our oracle. This subsection documents where such labels *exist*, so that the breadth and position analyses can, in principle, scale beyond the six-game core of Table S6. It reports label *availability and provenance* only. It does not report any score; the scores on the 44-game scored set drawn from it are in Section S1.

The counts are computed, not estimated. We intersect our bit-exact roster with the games that carry a label in the installed source of each project: the OCAtari object-property tables and the AtariARI `atari_dict`. OCAtari is preferred for render-aligned position labels because its RAM mode applies the render-time offsets that correct the

Table S7 Per-family applicability. What each method family requires of its target. "Intervenes" marks oracle-like do(u) access; "hypothesis" marks methods that receive a supplied circuit hypothesis to test. The raw-VCS column distinguishes analyses that read machine state directly from those that need a differentiable or layered model. A blank means the requirement does not apply.

Family	raw VCS	policy	gradients	layers/attn	intervenes	hypothesis
A neuroscience	yes	no	no	no	A2 only	no
B gradient	yes	no	yes	no	no	no
B intervention	yes	no	no	no	yes	no
C causal	yes	no	no	yes	yes	yes
C gradient (attr. patch)	yes	no	yes	yes	approx.	yes
C probing	yes	no	no	yes	no	no
C / A dim. reduction	yes	no	no	yes	no	no
oracle (control)	yes	no	no	yes	exact	—

Table S8 T3 semantic-label coverage. Availability of external position/object labels across the 64-game bit-exact roster, computed against the installed OCAtari object tables and the AtariARI `atari_dict`. This is a provenance record of which games *could* enter a label-dependent analysis; it reports no result on any of them. Every AtariARI-labeled game is also OCAtari-labeled, so OCAtari alone defines the maximal labeled set of 54.

Set	Count	Note
Our bit-exact roster	64	all Paper-1 games
OCAtari-labeled (offset-corrected)	54	largest labeled subset
AtariARI-labeled	22	all a subset of the OCAtari 54
Covered by both sources	22	richest (cross-checkable)
AtariARI-only	0	every AtariARI game is in OCAtari
No T3 label at all	10	excluded from label-dependent scoring

raw-RAM misalignment [33]; AtariARI labels are cross-checkable but do not carry those offsets. Table S8 summarizes the intersection. Of the 64 bit-exact games, 54 carry an OCAtari label, which is the maximal labeled subset: every one of the 22 AtariARI-labeled games is also in the OCAtari 54, so the AtariARI-only count is 0 and OCAtari alone fixes the ceiling. The 22 games covered by both sources are the richest and the only cross-checkable ones. Ten games carry no T3 label from either source and are excluded from any label-dependent scoring. In percent of the roster, OCAtari labels 84%, both sources cover 34%, and 16% have no label at all.

For traceability we enumerate the sets. The 54 OCAtari-labeled games are: `air_raid`, `alien`, `amidar`, `assault`, `asterix`, `asteroids`, `atlantis`, `bank_heist`, `battle_zone`, `beam_rider`, `berzerk`, `bowling`, `boxing`, `breakout`, `carnival`, `centipede`, `chopper_command`, `crazy_climber`, `demon_attack`, `double_dunk`, `enduro`, `fishing_derby`, `freeway`, `frostbite`, `gopher`, `hero`, `ice_hockey`, `jamesbond`, `kangaroo`, `krull`, `kung_fu_master`, `montezuma_revenge`, `ms_pacman`, `name_this_game`, `pacman`, `phoenix`, `pitfall`, `pong`, `pooyan`, `private_eye`, `qbert`, `riverraid`, `road_runner`, `robotank`, `seaquest`, `skiing`, `space_invaders`, `star_gunner`, `tennis`, `time_pilot`, `up_n_down`, `venture`, `video_pinball`, and `yars_revenge`. The 22 games covered by both sources are: `asteroids`, `berzerk`, `bowling`, `boxing`, `breakout`, `demon_attack`, `freeway`, `frostbite`, `hero`, `montezuma_revenge`, `ms_pacman`, `pitfall`,

pong, private_eye, qbert, riverraid, seaquest, space_invaders, tennis, venture, video_pinball, and yars_revenge. The 10 games with no T3 label, dropped from any label-dependent scoring, are: defender, elevator_action, gravitar, journey_escape, solaris, surround, tutankham, videochess, wizard_of_wor, and zaxxon.

S3.2 Scored set: the 44-game applicable subset

The T3-coverage of Section S3.1 sets the outer breadth of the sweep, but not every labeled game can measure faithfulness. The interpretability methods are scored on 44 games, not on all 54 T3-labeled games of Table S8. We hold out 10 games for two methodological reasons, decided before scoring. This is a scope decision about where the benchmark can measure anything, not a result. The excluded games keep their exact T1 and T2 ground truth on record; only their score cells are held out.

The first reason is a lack of causal structure at the shared analysis state. Six games fail the oracle cause-density gate: fewer than four candidate causes move the shared output above the floor ($|\Delta y| > 0.5$). A frame with almost no true causal signal cannot measure faithfulness, and running methods there would only add noise to the leaderboard. These six are crazy_climber, pooyan, time_pilot, asteroids, battle_zone, and star_gunner. The second reason is the absence of a moving target for a position method. Seven games carry no verified label on a moving sprite (their verified labels, if any, are static, such as score or life counters), so no sprite moves under intervention and a position method has nothing to recover. These seven are amidar, beam_rider, robotank, up_n_down, asteroids, battle_zone, and star_gunner. The two criteria overlap for exactly three games, namely asteroids, battle_zone, and star_gunner, which fail both. Counting each excluded game once, the union is 3 gate-only plus 4 static-only plus 3 that fail both, which is 10, so the scored set is $54-10=44$ games. Table S9 lists the 10 with their reason.

Each of the 44 scored games both clears the cause-density gate and carries a sprite that moves under intervention. Hence every one of them contributes to the all-regime leaderboard and to the position regime. The six core games of Table S6 are the subset of these 44 that also carry the full tiered ground truth and the exact oracle.

The analysed streams are bit-exact to frame 105.

Paper 1 certifies the port over 30 RAM and 60 screen frames, while the shared testbed scores at frame 105. We therefore replayed, for each of the 44 scored games, the exact seed-0 testbed stream through the boot and step code the methods use, and compared it with the seed-patched xitari reference on every frame from boot to frame 105 (tools tools/xai_study/repro/p2_stream_{dump.jl,xitari.sh,check.py}, result table p2_stream_check.md). All 44 games match on every frame, in all 128 RAM bytes and in every screen pixel. A negative control (Pitfall replayed without its start-up input) diverges at the boot frame, so the comparison is sensitive. Each game boots with its own ROM settings, the map Paper 1's conformance suite uses. An earlier version of the runners booted 37 games with generic settings, which skipped the start-up inputs of seven games then in the scored set (air_raid, beam_rider, double_dunk, gopher, pitfall, private_eye, yars_revenge) and used a different screen crop for

Table S9 Games held out from the scored set. The 10 games excluded from the 44-game scored set, with the methodological reason for each. "Gate" marks a game that fails the oracle cause-density gate (fewer than four causes with $|\Delta y| > 0.5$ at the shared analysis state). "Static" marks a game with no verified label on a moving sprite (its verified labels, if any, are static, e.g. score or life counters), so no sprite moves under intervention for a position method to recover. The two criteria overlap for three games marked "gate + static", so counting each game once the union is 3 gate-only plus 4 static-only plus 3 both, which is 10, and the scored set is $54 - 10 = 44$. These games keep their exact T1/T2 ground truth (Section S3.1); only their score cells are held out.

Game	Reason	Why held out
crazy_climber	gate	too few causes clear the density floor
pooyan	gate	too few causes clear the density floor
time_pilot	gate	too few causes clear the density floor
amidar	static	no verified label; no sprite moves under intervention
beam_rider	static	verified labels static; no sprite moves under intervention
robotank	static	no verified label; no sprite moves under intervention
up_n_down	static	no verified label; no sprite moves under intervention
asteroids	gate + static	too few causes clear the density floor and no sprite moves under intervention
battle_zone	gate + static	too few causes clear the density floor and no sprite moves under intervention
star_gunner	gate + static	too few causes clear the density floor and no sprite moves under intervention

three (`air_raid`, `carnival`, `pacman`). Those earlier streams were bit-exact as well (RAM 42/42 against `xitari` without start-up inputs), but they did not start where the Arcade Learning Environment starts. All reported scores use the corrected boot, and the scored set itself was re-derived under it: the cause-density gate and the verified sprite labels were recomputed, which adds `asterix`, `enduro` and `skiing` and removes `beam_rider` (no sprite moves under intervention at the shared state), for 44 scored games.

The full per-game roster is in Table S10. It lists all 64 bit-exact games with their label source, the number of causally verified T3 labels, whether the game carries a moving tracked position, and whether it enters the scored set (with the held-out reason otherwise). It makes the $64 \to 54 \to 44$ funnel explicit at the level of individual games: 64 in the roster, 54 with a T3 label, and 44 in the scored set. This mirrors the per-game "Ground Truth ROMs" page of the public dashboard.

S3.3 Per-game ground-truth roster

Table S10 gives the full per-game roster behind the counts above, one row for each of the 64 bit-exact games. It mirrors the "Ground Truth ROMs" page of the public dashboard. For each game it records the T3 label source (OCAtari, both OCAtari and AtariARI, or none), the number of causally verified T3 labels, whether the game carries a moving tracked position, and whether it enters the 44-game scored set. Games that are held out show the reason in the last column. Reading down the last column

makes the 64 → 54 → 44 funnel explicit: 64 games in the roster, 10 with no T3 label, 10 T3-labeled games held out from scoring for lack of causal structure or a moving target, and 44 in the scored set.

Table S10: **Per-game ground-truth roster.** All 64 bit-exact games, alphabetically. "Label source" is the T3 provenance (OCAtari, both sources, or — for no label). "T3 verified" is the count of causally verified T3 labels. "Moving pos." is Yes when the game carries a tracked sprite that moves under intervention. "Scored set" is Yes for the 44 scored games; otherwise it gives the held-out reason. The three held-out reasons match Table S9: the density gate, static-only labels, or both; "no T3 label" marks the 10 unlabeled games. This is a coverage record; no interpretability score appears in it.

Game	Label source	T3 verified	Moving pos.	Scored set
air_raid	OCAtari	10	Yes	Yes
alien	OCAtari	3	Yes	Yes
amidar	OCAtari	0	—	held out (static labels)
assault	OCAtari	14	Yes	Yes
asterix	OCAtari	7	Yes	Yes
asteroids	both	0	—	held out (gate + static)
atlantis	OCAtari	3	Yes	Yes
bank_heist	OCAtari	3	Yes	Yes
battle_zone	OCAtari	1	—	held out (gate + static)
beam_rider	OCAtari	2	—	held out (static labels)
berzerk	both	13	Yes	Yes
bowling	both	8	Yes	Yes
boxing	both	8	Yes	Yes
breakout	both	7	Yes	Yes
carnival	OCAtari	17	Yes	Yes
centipede	OCAtari	8	Yes	Yes
chopper_command	OCAtari	13	Yes	Yes
crazy_climber	OCAtari	2	Yes	held out (density gate)
defender	—	0	—	no T3 label
demon_attack	both	11	Yes	Yes
double_dunk	OCAtari	10	Yes	Yes
elevator_action	—	0	—	no T3 label
enduro	OCAtari	2	Yes	Yes
fishing_derby	OCAtari	7	Yes	Yes
freeway	both	14	Yes	Yes
frostbite	both	12	Yes	Yes
gopher	OCAtari	15	Yes	Yes
gravitar	—	0	—	no T3 label
hero	both	11	Yes	Yes
ice_hockey	OCAtari	10	Yes	Yes
jamesbond	OCAtari	5	Yes	Yes
journey_escape	—	0	—	no T3 label
kangaroo	OCAtari	9	Yes	Yes
krull	OCAtari	11	Yes	Yes
kung_fu_master	OCAtari	5	Yes	Yes
montezuma_revenge	both	15	Yes	Yes
ms_pacman	both	15	Yes	Yes
name_this_game	OCAtari	4	Yes	Yes
pacman	OCAtari	8	Yes	Yes
phoenix	OCAtari	8	Yes	Yes
pitfall	both	10	Yes	Yes
pong	both	6	Yes	Yes
pooyan	OCAtari	4	Yes	held out (density gate)
private_eye	both	12	Yes	Yes
qbert	both	6	Yes	Yes
riverraid	both	15	Yes	Yes
road_runner	OCAtari	7	Yes	Yes
robotank	OCAtari	0	—	held out (static labels)
seaquest	both	11	Yes	Yes
skiing	OCAtari	8	Yes	Yes
solaris	—	0	—	no T3 label
space_invaders	both	8	Yes	Yes

Table S10 (continued)

Game	Label source	T3 verified	Moving pos.	Scored set
star_gunner	OCAtari	0	—	held out (gate + static)
surround	—	0	—	no T3 label
tennis	both	11	Yes	Yes
time_pilot	OCAtari	2	Yes	held out (density gate)
tutankham	—	0	—	no T3 label
up_n_down	OCAtari	0	—	held out (static labels)
venture	both	16	Yes	Yes
video_pinball	both	7	Yes	Yes
videochess	—	0	—	no T3 label
wizard_of_wor	—	0	—	no T3 label
yars_revenge	both	4	Yes	Yes
zaxxon	—	0	—	no T3 label

S4 Claims and evidence

Every headline claim in the paper rests on a specific artifact. The table in this section makes that dependency checkable. For each claim we list the figure or table that supports it, the games and methods it uses, the metric, the artifact it traces to, and the limitation that bounds it. Faithfulness, sufficiency, and minimality are the three axes of Section 3.4. The numbers themselves are sourced in Section S5. The table below names which evidence supports which claim; it does not repeat the values. All aggregate scores are taken over the 44-game scored set of Section S3.2. One split matters for how the table is read. The all-regime faithfulness gap between causal/intervention and gradient/correlational methods is 0.324 (causal/intervention mean 0.709 vs. gradient/correlational 0.384), and it is the primary evidence for the headline. The position-only gap is now significant on this set: bootstrapping the family-mean gap over the 44 games gives a mean of 0.338 with a 95% interval of $[0.293, 0.383]$ that excludes zero. At the earlier six-game scope this gap was 0.238 with an interval of $[-0.049, 0.315]$ that crossed zero; scaling to 44 games resolves it into a significant result.

S5 Number provenance

Every headline number traces to a committed record, the script that produced it, and the command that re-derives it. This section is the in-paper copy of that traceability table. It mirrors Section 6.5 of the reproducibility document (`xai_2_interpretability/REPRODUCIBILITY.md`). The values are the measured, committed ones at the current revision. The position-regime gap and its 95% interval are read from `tools/xai_study/compare/out/position_bootstrap.json`; the other aggregates from `leaderboard.json`, rather than recomputed here. Each record also carries `seed`, `where`, `commit`, and `timestamp`, so the provenance of every value is self-documenting. The aggregate scores below are taken over the 44-game scored set of Section S3.2, not the six-game core; each per-method row is a mean over the games on which that method applies (`n_games` per method, e.g. 44 for most methods, 38 for ACDC and A1). The position-regime gap is now significant on this set.

One number needs a note, and we record it rather than hide it. On the 44-game scored set the sparse autoencoder's over-games mean $F = 0.1474$ (`triad_F`) and over-records mean 0.1474 (`faithfulness`) coincide, both read from the sparse-autoencoder

row of `leaderboard.json`; the figures and the main-text leaderboard plot this value (0.147). The finer object is the per-game Pong record $F = 0.041$ used in the named-not-causal separation of Section 1.4: it is a single-game illustration, not the leaderboard aggregate. No headline claim depends on which is read, since the sparse autoencoder sits near the floor under both.

S6 What this paper does not claim

The benchmark is sharp because its scope is narrow. Stating the boundaries keeps the result from being read wider than the evidence allows. We do not claim to recover a program's semantics from scratch. The study measures whether an explanation names the true causes and regenerates the behavior on a machine whose answer is already known. It does not show that any method discovers the meaning of a routine without that ground truth in hand. A faithful method recovers the causal structure but not the meaning of the variables, and this remaining gap is the point rather than a result to be explained away.

We do not evaluate learned agents. The subject throughout is the bit-exact VCS itself. The candidate causes are its registers, RAM cells, and program variables, not the weights of a trained policy. The neural-network parallels drawn in the discussion and in Fig. 5 are documented analogues from the literature, not claims proven on this benchmark. We label them as such, so that a reader does not take a mapped failure mode for a demonstrated one.

We do not prove that neural-network interpretability is impossible. The VCS gap is a necessary-condition stress test. A method that fails to recover the causal structure of a fully known machine cannot be trusted to recover it on an unknown one. Yet, a method that succeeds here is not thereby certified on networks. The "floor" framing is an evidence-transfer argument, not a proven lower bound on neural-network interpretability. Likewise, the end-to-end gradient control of Section S1 differentiates through the whole 15-frame analysis horizon only. We do not claim anything for longer horizons, and the leaderboard's gradient rows still use the one-step surrogate.

We do not measure human plausibility. The plausibility axis is a documented method-tradition proxy, computed from a fixed rubric rather than collected from raters, and we report it as a proxy throughout. Its job is to expose the danger zone where a convincing explanation is unfaithful, not to stand in for a human-subjects study. We also do not redistribute the ROMs. The substrate is validated against an external reference, and the records are reproducible from committed artifacts without the ROMs. The ROM provenance is given as hashes (Section S8), so a third party supplies their own legally obtained copies.

S7 Plausibility-proxy rubric and its sensitivity

The plausibility axis of the leaderboard is a documented, tradition-level proxy, not a human-subjects measurement. This section records how each value was assigned. It also shows that the headline contrast does not depend on the particular numbers chosen. The Methods (Section 3.4) name two reporting axes. Faithfulness is measured on the machine against the intervention oracle of Section 3.2. The plausibility

proxy stands in for how convincing a tradition's explanations are usually taken to be. We assign the proxy once per method tradition, before any faithfulness score on our machine is computed, so it cannot be tuned to the result. Its only job is to expose the danger zone of explanations that look convincing yet are unfaithful. For that job it must be coarse, fixed in advance, and sourced rather than precise.

The rubric assigns one proxy per tradition on a fixed five-step scale. It scores two documented properties of each tradition's typical output and takes their average. The first property is whether the explanation produces a dense, picture-like artifact that a reader sees as an account of the computation. Heatmaps and saliency-style maps over the input are read as convincing by construction. This is the perceptual-immediacy axis. The second property is whether the field has historically treated the tradition's output as a self-evident explanation rather than a hypothesis to be tested. This is the adoption-as-explanation axis. A tradition that scores high on both, such as gradient saliency, receives a high proxy. A tradition whose output is an austere intervention table or a circuit edge list, rarely mistaken for a finished story, receives a low one. The scale is deliberately blunt, $\{0.3, 0.5, 0.55, 0.75, 0.8, 0.85, 0.9\}$. A proxy meant to expose a danger zone gains nothing from false precision, and a finer scale would invite over-reading. Every value in Table S13 traces to the `plausibility_proxy` field of the per-method rows in `compare/out/leaderboard.json`. The table only groups those rows by tradition and records the rationale.

The paper draws a contrast between this proxy and measured faithfulness. On the committed records the two move in opposite directions. Across the thirty methods under test the Spearman rank correlation between faithfulness and the proxy is -0.50, a negative association. The traditions the field finds most convincing are, on this machine, among the least faithful. Twelve methods fall in the danger zone, defined here as a proxy of at least 0.70 together with a faithfulness below 0.40. They include the four correlational neuroscience analyses (A3–A6), the gradient methods (vanilla saliency, SmoothGrad, guided backprop, integrated gradients, expected gradients), linear probing, the NMF/PCA dictionaries, and the sparse autoencoder. The per-method extreme makes the point sharply. Vanilla saliency has the highest proxy, 0.9, yet scores faithfulness 0.092 on the position regime. Activation patching has a lower proxy, 0.5, and scores 1.0. Here the less faithful method is the more plausible one. These figures are produced by `tools/xai_study/compare/plausibility_sensitivity.py` from the leaderboard and recorded in `compare/out/plausibility_sensitivity.csv`.

A proxy fixed by a coarse rubric raises the question of whether the contrast is an artifact of the particular numbers. The answer is that it is not. We perturb the proxy under four deterministic schemes and recompute the relationship each time. The analysis uses fixed, index-keyed offsets rather than random draws, so it re-derives bit-for-bit. The first scheme adds a fixed per-method jitter of magnitude $\sigma \in \{0.05, 0.1, 0.2\}$, alternating in sign by row. The second compresses every proxy toward the common mean by a fraction σ, which preserves the rank order of traditions. The third uses the same index-keyed offsets at $\sigma \in \{0.1, 0.2\}$, large enough to swap neighbouring traditions' ranks. The fourth drops one whole tradition at a time and recomputes on the remainder. Across all fourteen perturbations the rank correlation between faithfulness and proxy stays negative. It ranges from -0.82 to -0.10. The danger zone stays populated,

with between seven and thirteen methods. The per-method anchor sign holds in every scheme where both anchor methods are present, the more plausible saliency remaining the less faithful. Two runs bound the range. The strongest is the leave-the-gradient-tradition-out run. It removes the most plausible and least faithful block at once, and still leaves seven danger-zone methods and a rank correlation of -0.82. The weakest is the leave-the-causal-tradition-out run, where the correlation falls to -0.10, close to zero; removing the faithful causal block flattens the association, so the contrast is carried in part by that block. Table S14 reports the full sweep. The contrast keeps its sign under every alternative proxy assignment we tried: plausibility and faithfulness are not the same axis, and high plausibility is no guarantee of faithfulness.

S8 ROM provenance: SHA-256 hashes

We do not redistribute the Atari 2600 game cartridges. They are third-party ROMs acquired through the `AutoROM` retrieval procedure, following the practice of the underlying emulator port [30]. To let a third party verify that they hold the exact binaries we ran against, we report a SHA-256 hash for each ROM in the bit-exact roster. The Data-availability statement points here.

Table S15 lists all 64 games of the Paper-1 bit-exact roster, so the hash record covers the six-game core (Table S6), the 44-game scored set (Section S3.2), and the wider 54-game labeled breadth set (Table S8). A dagger (†) marks the six core games. Each hash is computed over the raw cartridge image with `shasum -a 256`; reproducing it requires only the retrieved binary and a standard SHA-256 tool. The hashes fingerprint the exact inputs; they are provenance, not a result.

Table S15: **SHA-256 hashes of the 64 bit-exact ROMs.** One row per game of the Paper-1 roster. A dagger (†) marks the six core games; the 44 games of the scored set (Section S3.2) carry interpretability scores, and the remaining rows document the wider substrate whose ROM binaries are pinned but not scored. Hashes are computed with `shasum -a 256` over the retrieved cartridge image and are a provenance record only.

Game	SHA-256
air_raid	6b5a22ec074a5d870e0693d8717d2f7fbf559ba970995d75882f902bde1d7c9b
alien	9cd556c7d7f55aeb78fc5add45d74ce889c6e22cba6bb535ef1e8bd6b6c764c1
amidar	da5b760d3adaeba2cb8fc2367ae96e2b8add4da44cd47199b57ff4d54e682d9f
assault	d32ca4dd9e849af7b776bc73bf464468839ddab66d7e6efacf877fe6eb841dc2
asterix	b6161db0530d8445b5ef0e8b700500531e980aa10184e5196355f27a7ba60819
asteroids	d1a6e808412b4535c7e54fc25414943a9418b1095b4d4969dc33290f7eaa72a4
atlantis	cef4b7b91ea245f7827604a1c620a5ca7f94288c2c6b54d90a931de8eeb5227d
bank_heist	c32daffdb926afaa0b11948c85b0b059e26d29af5f9ce213c439727ccf3f3a8f
battle_zone	efd6f639e724a771a8331d7f408609d0dc0854533b3163b602720754267943ee
beam_rider	83b30cb52649ddf7a5dd6c3336722421aa435e81e0f85913a1011d81e4e6bb38
berzerk	cdff0422329de6bdf4ab59176e050fd3fa47d329bc0f065768c4d268838a5fd2

Table S15 (continued)

Game	SHA-256
bowling	dff44a85289f5e9f760254e4806d09038326cdfc42eba3e1cd7c64ed6e0b4dce
boxing	54760d7e071049fbef09370e136af270bddb5aa2a4fa9be0f404457e49f47377
breakout†	376323f051c3c373c887fd83abead39d87d844ff283d435f4addbfc1710c6fd5
carnival	80eab759b6bd4fb59e98d4db1295ad89b128a97d2e20ccb212c744ae642c2f76
centipede	6def669f54bd886f43026e7fc7f82b9acb92680bb3e3aa68da3767f8797c8cbe
chopper_command	055637282252b5378a2b7df573726e85afa91264ffb7243d23fab9518b3c207d
crazy_climber	486ad4cc89569d797dc5feff6e95d1d5f5f8e78331a4925ef3126bd2aa1bcf09
defender	e3c7ed7a073fe8f4b2711bebd624d0b249957daa0ebb93d78823fe36b235ba8c
demon_attack	8d72feeb267d23cb5d58bb4511566414884c572d2b4dcba8f6893ad7426442d4
double_dunk	c0ea53a91a396638662e2d37261407d6bc9d1b1f660e3101b2098d7ef2bc3190
elevator_action	f3b3ad03fd8cf9f348d31c1343580a7a9220d1f06c193165df277e47757a8846
enduro	ca9bb89755a6d6251fa0ad86983f088a53b4013853963cbeb80dca93466c99fc
fishing_derby	84753bc6ecb4dd9f47cd267f1fb15e16b18f7b2b54fbb4ea8207750009171cc1
freeway	3ef620234b98f22bfd7c06d2467bac9d208e763c9f4309452384d59a741ba893
frostbite	cbfdad89480def69d922adf16a1d89d55a6cb515929edf74e5bea884c9fb7834
gopher	b4aff03aeb0fb1c8f4914b5a329f51430f00a2d9350aba24573ad78032dd4697
gravitar	17a19cd7b5cae2b85f3a51fd10e92e141341f3e4238e9ea641cb97656a3ed0a0
hero	75bd02ec75458ea4bc2cba9b85bb281c3f9da106123e4600b054f0934e36b81f
ice_hockey	06df1fc568d9815cf6ed0d08b27555ee53819c210ee6c786aca9c61b1067b0a2
jamesbond	995253f0d3a2fac712c6ee3518f19fc5e6a8c800e2fd8dbfe49d45ad367a231f
journey_escape	963e2d3da52b6ccaff08c833b2d2cb0329a9b9d5f6b3265fcfd7e0830b914df0
kangaroo	a2826eaea3a854b63b4ca1d7423f1b3347311e0552c6b1f35065e50b2cbca523
krull	b40f10d14217093082b00b302bff8c6faa0d64c5221e9be36ea0ceb12477517a
kung_fu_master	3f6501a649ad83e970a25827bd492c56128c36535ae2c96f94bab39b27f939ac
montezuma_revenge	69d363a747549817599cace65572700dab9ea33f3a5f91979454fdfbeaa837c0
ms_pacman†	dde0b43c5dee7ce1a35ac8e2f625b32a16118512d9be4064115053b972280b14
name_this_game	cc219a7246e8f9454f3bd77345b388fbc2c7343e4ea0c4c864a2eb9b0481882a
pacman	58e781b472e0583b851b7e56cbfe87bd8294a291e6d39d49d7c999107ed5b918
phoenix	b01085bc5227a04b974846fe06b4018da84fd3e40ac6d1368f7f6bc07c336dfd
pitfall	c719b47714d87d17e05de770fab2ab0bdaf6248d411539d6ecdeddf7de3ae77b
pong†	41623e3c26148f96cbe77d56512845d224a63c8fe5bf90e38527155e39ec96d3
pooyan	ac9abbfc272deb0305451afa3d5cd58849684691cfa25cede01251fab36c131b
private_eye	c8f9ce1b2e804b6778dbc20da0c935b2af7d7e998ad372d44df66f16eb4b98ce
qbert†	3257221832a7607696b06b9985103c12a4c24c61b0c655e8f52aed8e38194b76
riverraid	4c6842b8af64fc75e599b773821b3e6bee8e5028f5d63414821b0d63f0b2dfe9
road_runner	4b6c2e68d6933b8bc79983f6a3bb11bda695b5dfbeeb8fd63d84406ba3f27b5a
robotank	c186409481e688401c8a7a9d5f0387ed448d18e297e287fd7c409752fa9a2a0d
seaquest†	fbc29f4678f69ac27fe46da298c8c22f98cba2a3e5491e6b308fd9bf68d2ee43
skiing	804eff5f0196375b33614aa18cb7f2f883681ced1654e5e8ca702204756a19bb
solaris	0afa36e5f4d8d77e38716673c96cda3ad11d421283f5a5ae77ec7a01e7718a4e
space_invaders†	7224b17462b992d67f4e06a3c85f269c9822b06df6015bf038b55f384ced0301
star_gunner	3ad501a39280de82bd604c6d8ae915fe93c1e6ebfd97685ab097c5dbef9822c3

Table S15 (continued)

Game	SHA-256
surround	c6864f5c9f43f0590226b4f2943d00fb8a90a2c6f799963de75016d5c07affc3
tennis	5a2052f020bfb19f5fa4db8a3149287a875bc4025059dc66b747451fb7fe4909
time_pilot	8ea97e335a2fb122faa591a86e695de1eddb88e9370d3ead6e82d308f6e81c7d
tutankham	403397fe85838ebc94bf063e63756b7656b9c962b9c6392b991d37ef4ad4a3d4
up_n_down	74890a43e59134ea88e0a3aa6cae91615299cc41d74a464e0e31070d5d36062c
venture	1870b0dd6fb1de45e44e1ad114c8f558104351cfb11495dc4fa5c7b0fc58f455
video_pinball	cef82c39bbb035f1ab5b900548de410371753ff717f3bf69a2e314cbd55fe937
videochess	d44f2a115393b0a6cbcd5398d62369e652f3670a9ee585883d183e740edad0f7
wizard_of_wor	d69ee89e65c34cd3152d98d778de913dd7f59e7309f2bcbd3ebead585065e747
yars_revenge	ff777c8d4ea06569156fcb2f5ba0e00eebaf2c2c509e5829109e5aefed199c66
zaxxon	e1c323ce00cccdd4bc737fd44abcef35b24fee04fd90bf9c9374e18b3b2ae6bf

Table S11 Claims and their evidence. Each headline claim maps to the figure or table that supports it, the games and methods involved, the metric, the committed artifact, and the limitation that bounds the claim. The artifact column points to the record that Section S5 traces to a value. The limitation column states the scope, so the claim is not read wider than the evidence.

Claim	Evidence	Methods / games	Metric	Limitation
Causal/intervention methods beat gradient/correlational on faithfulness across all regimes	Fig. 2; leaderboard	25 methods, 44 games	F mean (gap 0.324)	robust: causal/interv. 0.709 vs. gradient/corr. 0.384; this all-regime gap is the primary evidence
The gap holds with the same sign on the discrete position regime	Fig. 2; `position_bootstrap`	13 rows, 44 games	F position (gap 0.338)	significant on the 44-game set: family-mean gap 0.338 $[0.293, 0.383]$ excludes zero (crossed zero at 6 games)
A faithful causal method reaches the oracle ceiling where the default tool stays low	`faithful_demo`; Fig. 2	act. patching vs. vanilla saliency, 44 games	F 1.0 vs. 0.092 (position regime; all-regimes saliency F 0.361, plotted in Fig. 2)	one decisive pair
The naive gradient of a discrete position output is provably zero	Prop. 1 (Section 3.2); `oracle_grad`	all gradient methods	grad@position = 0; sampler restores 166	position path only; restored gradient is low/mixed
High plausibility coexists with low faithfulness (the danger zone)	Fig. 2; `faithful_demo`	vanilla saliency	proxy 0.9, F 0.092 (position regime; all-regimes F 0.361, plotted in Fig. 2)	proxy, not measured plausibility
Faithful attribution recovers the wiring but not the computation's semantics	Fig. 4 (Section S3); discussion	Phase B vs. C	F vs. S, M	necessary-condition stress test
A gradient through the whole program gives the same map as the one-step surrogate on most games, and no higher faithfulness on average	Section 1.2; `e2e_core_summary.json`, `e2e_vs_single.json`	vanilla saliency, Grad×Input; 44 scored games	F content, F position	control; not a leaderboard row
The failure modes have documented NN analogues	Fig. 5	taxonomy	—	analogues, not proof

Table S12 Headline-number provenance. Each number → its committed record and JSON key → the generating or aggregating script → the command that reproduces it. Values are the measured, committed numbers at this revision (consistent with REPRODUCIBILITY Section 6.5); the position-regime gap CI is read from `position_bootstrap.json`, all other aggregates from `leaderboard.json`. Paths are relative to `tools/xai_study/`.

Number	Value	Record (JSON key)	Command
Faithfulness gap, all regimes (causal − gradient)	0.324 [0.297, 0.351], excl. 0	`position_bootstrap.json`: `all_regime.family_mean_gap`, `ci95`	`position_bootstrap.py`
Causal/interv. mean F, all regimes	0.709 n=11	`leaderboard.json`: `headline_contrast.all_regimes.causal_...mean`	`leaderboard.py`
Gradient/corr. mean F, all regimes	0.384 n=14	`leaderboard.json`: `headline_contrast.all_regimes.gradient_...mean`	`leaderboard.py`
Faithfulness gap, position regime (bootstrap over games)	0.3381 [0.293, 0.383], excl. 0	`position_bootstrap.json`: `position.family_mean_gap`, `ci95`	`position_bootstrap.py`
Causal/interv. mean F, position regime	0.560 ±0.3214, n=4	`leaderboard.json`: `headline_contrast.position_regime.causal_...mean`	`leaderboard.py`
Gradient/corr. mean F, position regime	0.2221 ±0.157, n=9	`leaderboard.json`: `headline_contrast.position_regime.gradient_...mean`	`leaderboard.py`
Decisive pair, position regime (act. patch vs. vanilla sal.)	1.000 vs. 0.092	`faithful_demo`: `pair_contrast`	`faithful_demo.py`
Plausibility proxy, decisive pair	0.5 vs. 0.9	`faithful_demo`: `..._method.plausibility_proxy`	`faithful_demo.py`
ACDC sufficiency S	0.314	`leaderboard.json`: ACDC `triad_S`	`leaderboard.py`
Whole-state minimality M	0.0746	`leaderboard.json`: A8 `triad_M`	`leaderboard.py`
Methods scored (rows)	31	`leaderboard`: `n_methods`	`leaderboard.py`
Per-game records aggregated	1862	`leaderboard`: `n_per_game_records_aggregated`	`leaderboard.py`
Naive gradient, discrete position output	position 0; content 1.0; sampler 166	`oracle_grad_pong.json`: `value`, `extra.index`, `extra.sampler`	`oracle_grad.jl`
Exact oracle max $\lvert\Delta\text{score}\rvert$ (Pong)	17.0	`oracle_pong_score.json`: `value`	`oracle_intervene.jl`

Scripts live under `tools/xai_study/compare/` (Python) and `tools/xai_study/ground_truth/` (Julia); records under the corresponding `out/` trees. JSON keys are abbreviated with "..." where nested; the full paths are in REPRODUCIBILITY Section 6.5.

Table S13 Plausibility-proxy rubric. The proxy assigned to each method tradition, with the documented rationale. Values are the `plausibility_proxy` field of the leaderboard rows (`compare/out/leaderboard.json`). They are tradition-level and fixed before any faithfulness score is computed, so the proxy cannot be tuned to the measured faithfulness. A higher value means an explanation that the field tends to find more convincing, not one that is more faithful. The oracle (Section 3.2) is the positive control and is not a method under test.

Tradition	Proxy	Rationale (documented basis)
gradient	0.90	Dense input-space heatmaps read as a visual account of the decision; the most widely adopted XAI outputs, routinely shown as if self-explanatory.
probing	0.85	A trained decoder that "finds" a concept reads as evidence the concept is used; high adoption, though decodability is not causal use.
correlational	0.80	Tuning curves and coupling spectra match the neuroscience idiom of a convincing single-unit account.
dim. reduction	0.75	Low-dimensional factors and dictionary atoms look like a clean summary of the representation.
intervention	0.55	Occlusion and counterfactual outputs are quantitative tables, less immediately picture-like, treated as tests rather than stories.
causal	0.50	Circuit edge lists and patching results are austere artifacts rarely mistaken for a finished narrative.
descriptive	0.30	Whole-state recording is an explicit non-explanation: it withholds nothing and reads as a dump, not an account.
oracle (control)	1.00	The Section 3.2 ground truth by construction; positive control, not a method under test.

Table S14 Proxy-sensitivity sweep. The faithfulness-versus-proxy relationship under each perturbation, from `compare/out/plausibility_sensitivity.csv`. "ρ" is the Spearman rank correlation between faithfulness and proxy over the methods present. "danger" is the count of high-proxy (≥ 0.70), low-faithfulness (< 0.40) methods. "anchor" is 1 when the more plausible saliency is the less faithful of the saliency/activation-patching pair, and blank where a leave-one-out run removes an anchor. The baseline rubric is the first row. Across every scheme the correlation stays negative and the danger zone stays populated.

Scheme	Param	n	ρ	Danger	Anchor
baseline	—	30	−0.50	12	1
additive jitter	$\sigma = 0.05$	30	−0.52	12	1
additive jitter	$\sigma = 0.10$	30	−0.53	12	1
additive jitter	$\sigma = 0.20$	30	−0.56	13	1
rank-preserving	$\sigma = 0.10$	30	−0.50	12	1
rank-preserving	$\sigma = 0.20$	30	−0.50	12	1
rank-jittering	$\sigma = 0.10$	30	−0.53	12	1
rank-jittering	$\sigma = 0.20$	30	−0.56	13	1
leave-out causal	—	24	−0.10	12	
leave-out correlational	—	26	−0.58	8	1
leave-out descriptive	—	29	−0.45	12	1
leave-out dim. red.	—	27	−0.55	10	1
leave-out gradient	—	20	−0.82	7	
leave-out intervention	—	25	−0.42	12	1
leave-out probing	—	29	−0.52	11	1